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Mouse germline cysts contain a fusome that mediates oocyte development

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This manuscript provides evidence that mouse germline cysts develop an asymmetric Golgi, ER, and microtubule-associated structure that resembles the fusome in *Drosophila* germline cysts. This **fundamental** study provides new evidence that fusome-like structures exist in germ cell cysts across species. Overall, the data are **convincing** and represent a significant advance in our understanding of germ cell biology.

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Abstract

Mouse female primordial germ cells (PGCs) undergo five synchronous, incomplete mitotic divisions, and send each resulting germline cyst into meiosis to fragment and produce 4-6 oocytes and 24-26 supportive nurse cells. However, no system of polarity has been found to specify mammalian oocytes, link them appropriately to nurse cells and enable them to acquire high quality organelles and cytoplasm. We report that mouse cysts develop an asymmetric Golgi, endoplasmic reticulum (ER) and microtubule-associated “fusome”, similar to the oocyte-determining fusome in *Drosophila* cysts. The mouse fusome distributes asymmetrically among cyst cells and enriches in future oocytes with Pard3, and Golgi-endosomal UPR (unfolded protein response) proteins. Spindle remnants rich in stable acetylated microtubules, like those building the *Drosophila* and *Xenopus* fusomes, transiently link early mouse cyst cells for part of each cell cycle. A non-random gap in these microtubules predicts that initial cysts fragment into similar six-cell derivatives, providing a potential mechanism for producing uniform oocytes. Together with previous studies these results argue that a polarized fusome underlies the development of female gametes from the PGC to follicular oocyte stages in diverse animals including mammals.

Introduction

Early female germ cell development occurs within interconnected cysts formed by consecutive incomplete divisions of a germline stem cell (GSC) or primordial germ cell (PGC) in diverse animals. These include basal species like hydra ([Littlefield, 1994](#) ) , nematodes like *C. elegans*, insects like *Drosophila*, and vertebrates including fish, frogs, mice and humans ([Gondos, 1973](#) ; [King, 1970](#) ; [Lin et al. 1994](#) ; [Pepling and Spradling, 1998](#) ; [Kloc et al. 2004](#) ; [Marlow and Mullins, 2008](#) ; review, [Spradling, 2024](#) ). As they develop, *Drosophila* and other insect cyst cells become interconnected by ring canals and acquire polarity associated with a microtubule-rich “fusome” ([Giardina, 1901](#) ; [King, 1970](#) ; [Lin et al. 1994](#) ; [Büning, 1994](#) ; [Huynh and St Johnston, 2004](#) ). In *Drosophila*, the fusome arises from mitotic spindle remnants generated during incomplete cytokinesis, contains ER cisternae and displays Par gene and microtubule-dependent polarity that is essential for oocyte production. Later, in meiotic prophase, the fusome mediates the movement of centrioles, mitochondria, and germ plasm regulators such as oskar mRNA into the oocyte where they form the Balbiani body (Bb) ([Mahowald and Strassheim, 1970](#) 

Cox and Spradling, 2003 [\[1\]](#)). Remarkably, mammalian oocytes undergo these same events (review: Spradling et al. 2022 [\[2\]](#)), but a fusome has not been detected, raising the possibility that mammalian cyst polarity and oocyte specification use a different system.

Studies in *Xenopus* long provided suggestions that cyst polarization in at least some vertebrate species resemble the invertebrate process (Kloc et al. 2004 [\[3\]](#); review: Huynh and St Johnston, 2004 [\[4\]](#)). Recently a detailed study using modern methods characterized fusome-containing *Xenopus* cysts sharing many such similarities including production of relatively few oocytes and mostly nurse cells (Davidian and Spradling, 2025 [\[5\]](#)). Here, we report new studies on mouse cyst polarization. In the mouse, germ cells are induced around embryonic day 6.5 (E6.5) (Lawson et al. 1999 [\[6\]](#)) and begin migrating toward the gonad as PGCs. En route, PGCs start to reprogram their epigenetic state to pluripotency (Seki et al. 2007 [\[7\]](#); Saitou et al. 2011 [\[8\]](#); Loda et al. 2022 [\[9\]](#); Liu et al. 2025 [\[10\]](#)). After they enter the gonad at E10.5, PGCs take on germ cell character (Nicholls et al. 2019 [\[11\]](#)), turn on DDX4 and initiate cyst formation while reprogramming is still ongoing. During E11.5 they induce *Dazl*, a major post-transcriptional mediator of early germ cell development, pluripotency re-establishment, and meiotic entry (Ruggiu et al. 1997 [\[12\]](#); Haston et al. 2009 [\[13\]](#); Gill et al. 2011 [\[14\]](#); reviewed in Fu et al. 2015 [\[15\]](#); Zagore et al. 2018 [\[16\]](#); Yang et al. 2020 [\[17\]](#)). Even before meiotic entry, *Dazl* is needed to form stable cysts in zebrafish (Bertho et al. 2021 [\[18\]](#)) and in mice (Rosario et al. 2019 [\[19\]](#)).

The formation of germline cysts in mice is more complicated than in most animals. Mouse germline cysts arise from five rounds of mitotic divisions beginning at E10.5, so that each PGC generates almost 32 cells. However, the cysts break variably into an average of 4.8 smaller cysts before these divisions are completed at E14.5 when cells enter meiosis (Lei and Spradling, 2013 [\[20\]](#); Levy et al. 2024 [\[21\]](#)). During meiosis, each smaller cyst slowly loses component cells, as every 2 days about one cell on average becomes activated to act as a nurse cell. The nurse cell transfers most of its cytoplasm into the residual cyst, shrinks to a small remnant and undergoes programmed cell death (Lei et al. 2016 [\[22\]](#); Niu and Spradling, 2022 [\[23\]](#); Ikami et al. 2023 [\[24\]](#)). In the pachytene substage of meiosis I, there is a substantial reorganization that brings mitochondria and ER sheets into close proximity by E18.5 (Ruby et al., 1969 [\[25\]](#); Nogawa et al. 1988 [\[26\]](#); Pepling and Spradling, 2001 [\[27\]](#)). Cyst breakdown is complete by 5 days after birth (P5) with the formation of 4-6 primordial follicles, each containing a single oocyte in which organelles largely transferred from nurse cells are gathered into an aggregate known as the Balbiani body (Niu and Spradling, 2022 [\[23\]](#)). Cyst polarity guides cyst fragmentation and oocyte specification, polarization and organelle acquisition. Despite this, the only prior hint of a mouse fusome was a round structure that could be stained with certain antisera in PGCs and 2-cell cysts (Hahnel and Eddy, 1987 [\[28\]](#); Pepling and Spradling, 1998 [\[29\]](#)).

Germline cysts also support germ cell rejuvenation, the universal process that allows damage to be removed each generation so that species can persist without known limit. Germline rejuvenation takes place during meiosis in single celled eukaryotes (Unal et al. 2011 [\[30\]](#); Suda et al. 2024 [\[31\]](#); Xiao and Unal, 2025 [\[32\]](#)). Processes that rejuvenate organelles such as mitochondria, restore rDNA copy number, and reset epigenetic chromatin marks have also been documented in meiosis during animal gametogenesis (Cox and Spradling, 2003 [\[1\]](#); Hill et al. 2014 [\[33\]](#); Reik and Surani, 2015 [\[34\]](#); Bohnert and Kenyon, 2017 [\[35\]](#); Lieber et al. 2019 [\[36\]](#); Pang et al. 2023 [\[37\]](#); Yamashita, 2023 [\[38\]](#); Xiao and Unal, 2025 [\[32\]](#); Spradling et al. 2025 [\[5\]](#)). The Balbiani body in newly formed oocytes is associated with germ plasm formation and selection for mitochondrial function in many species (Heasman et al. 1984 [\[39\]](#); Kloc et al. 2004b [\[40\]](#); Marlow and Mullins, 2008 [\[41\]](#); Spradling et al. 2022 [\[23\]](#); Sekula et al. 2024 [\[42\]](#)). Currently, however, germline rejuvenation events are thought to skew toward late oogenesis (Bohnert and Kenyon, 2017 [\[35\]](#); Xiao and Unal, 2025 [\[32\]](#)).

Here we report that mouse PGCs and cyst cells elaborate a fusome-like structure that is rich in Golgi, endosomal vesicles and ER associations. The mouse fusome distributes asymmetrically with stable microtubules that persist for part of the cell cycle. By the 8-cell stage, cyst cells reorganize into a rosette configuration that brings cells with multiple bridges (potential oocytes) and associated fusome material closer together. Developing cysts at this stage usually fragment non-randomly at sites lacking microtubule connections into six-cell cysts, implying that mouse ovarian

cyst structure is more highly controlled than previously realized (Lei and Spradling, 2013 [DOI](#); Levy et al. 2024 [DOI](#)). Beginning as PGCs and throughout cyst formation, endosome-Golgi associated degradation (EGAD)-mediated UPR pathway proteins including Xbp1 are expressed, suggesting that the oocyte proteome quality control begins very early and continues. After pachytene, centrosome/Golgi-rich elements move to the oocyte and mediate the acquisition of organelles by the Bb. These results imply that a polarized fusome whose properties have been significantly conserved in diverse animals including mammals underlies female gamete development from the PGC to follicular oocyte stages.

Results

PGCs contain a Golgi-rich structure known as the EMA granule

Early observations of a germ cell “(e)mbryonic (m)ouse (a)ntigen (EMA)” (Hahnel and Eddy, 1987 [DOI](#)) identified an “EMA granule” in mouse primordial germ cells that resembles the *Drosophila* spectrosome, a fusome precursor (Lin et al. 1994 [DOI](#); Pepling and Spradling, 1998 [DOI](#)). To investigate this relationship further, we determined that the EMA granule appears in PGCs as early as E9.5 and continues to be expressed in later cysts (Figure 1A [DOI](#), Figure S1A [DOI](#)). We used lineage labeling to mark cysts (Video S1B) derived from single PGCs, and 3D surface rendering and volume quantification (Figure S1C), to demonstrate a significantly uneven distribution of the EMA granule within PGC daughter cells (Figure 1B [DOI](#)). This asymmetry and studies described below prompted us to name the EMA-rich structure the “mouse fusome”.

The EMA-rich material reorganized and clustered together in female cysts by E13.5 (Figure 1C [DOI](#); Video S1D1-2 [DOI](#); Figure S1A [DOI](#)), whereas in male cysts it remained separate and similar-sized in each cell (Figure 1C' and graph [DOI](#); Figure S1E' [DOI](#)) reminiscent of male differences in the *Drosophila* fusome (Diegmiller et al. 2023 [DOI](#)). In females, cyst regions with enriched fusome staining contained an average of 3 Tex14-positive bridges, whereas unenriched zones had an average of just 1 (Figure 1D-D' graph [DOI](#); Figure S1F [DOI](#)). Thus, highly branched cells with more bridges, which are known to give rise to most oocytes (Lei and Spradling, 2013 [DOI](#); Lei and Spradling, 2016 [DOI](#); Ikami et al. 2023 [DOI](#)), contain more fusome. Similar clustering, known as rosette formation (Figure 1H [DOI](#)), and increased fusome accumulation in pro-oocytes, characteristically occur as germline cysts grow in *Drosophila* and other species (see Hegner, 1914 [DOI](#); Büning, 1994 [DOI](#); Huynh and St Johnston, 2004 [DOI](#)). Rosette formation is thought to prepare for sharing between nurse cells and oocytes.

The mouse fusome is enriched in Golgi elements and vesicles

The fusome co-stained with the Golgi protein Gm130 (Figure 1F [DOI](#), Figure S1H [DOI](#)) and the recycling endosomal protein Rab11a1 (Figure 1F' [DOI](#)). Electron microscopy (EM) of mouse germ cells showed a prominent Golgi and endocytic vesicles in most E11.5 and E12.5 cyst germ cells in a subregion of similar shape to the EMA granule in light micrographs. (Figure 1E' [DOI](#); Figure S1K-L [DOI](#)) consistent with a previous EM observation (Jeon and Kennedy, 1973 [DOI](#)). We found evidence that EMA granules require ongoing Golgi activity to be maintained. In E11.5 mouse ovaries cultured in vitro, treatment with the Golgi-inhibitor Brefeldin A for 6 hours caused EMA granules to shrink, but many reformed within 18h of its removal (Videos S1I1-3). The fusome often partly overlaps with ER (Figure S1K-K' [DOI](#)). EM sections of E14.5 cysts revealed clusters of Golgi within adjacent cyst cells near ring canals as expected for cysts after rosette formation (Figure 1E, 1E'' [DOI](#)). EMA-1 antisera react with multiple fucosylated glycolipids (Apostolopoulos et al. 2015 [DOI](#); Figure S1J [DOI](#)). EMA staining disappears from germ cells at E14.5 (Figure 1I [DOI](#)), however, very similar (but non-germ cell-specific) staining is also seen with wheat germ agglutinin (WGA) (Figure 1G [DOI](#)) and this staining continues at E13.5 and later stages (Figure 1G' [DOI](#); Figure S1G [DOI](#)).

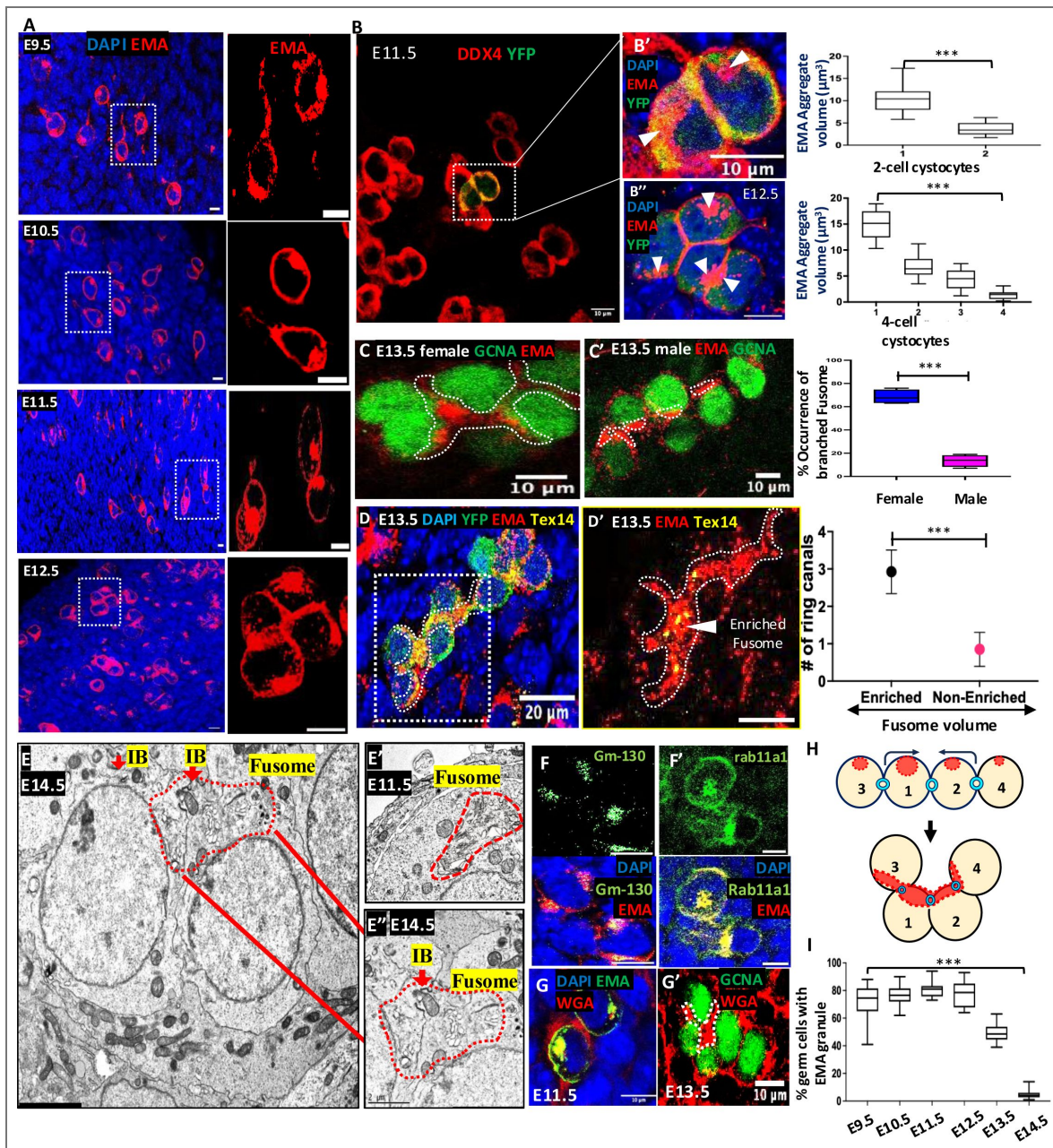


Figure 1. Mouse pre-meiotic PGCs contain a “fusome.”

A. PGCs and early germline cysts from E9.5-E12.5 ovaries. EMA (red), DAPI (blue). Boxed regions magnified at right (R) (arrows, EMA granules). B. EMA granule asymmetry in a 2-cell E11.5 cyst: Yellow cells represent a lineage-labelled 2-cell cyst marked with both YFP (green, lineage) and DDX4 (red). B'. Boxed region showing EMA granules (white triangles). Graph at R: EMA granule volumes consistently differ between daughter cells in 2-cell cysts. N=16. B". Varying volumes of daughter cells within E12.5 4-cell lineage-labeled cysts. EMA granules (white triangles), EMA (red). Graph at R: EMA volume asymmetry, N=18; C. Rosette formation in E13.5 ovary and E13.5 testis (C'). GCNA (nuclei, green), EMA (fusome, red; outline, dotted white). Graph at R: % of female (blue) and male (red) cysts with branched fusomes indicative of rosette formation (N=26 for each). D. Ring canal enrichment in high-fusome cells. A E13.5 lineage-labeled (YFP, green) cyst, fusome (EMA, red; outline, dotted white), and ring canals (TEX14, yellow). D' Zoomed image (boxed region in D) showing branched region with enriched fusome (white triangle) containing multiple ring canals. Graph at R: Ring canal number vs. fusome enrichment ($\geq 10 \mu\text{m}^3$). N=54. E-E". EM of an E14.5 cyst in rosette configuration showing a Golgi-rich fusome spanning an intercellular bridge (E'). E'. EM of an E11.5 PGC with a Golgi-enriched region (red outline) and likely EMA granule (compare to 1A-B). F-F'. E11.5 germ cells with EMA granules (EMA, red) co-stained with the Golgi markers F. (GM130, green) or F' (Rab11a1, green). G. Co-staining of WGA (red) and EMA (green) in E11.5 germ cells. G'. WGA (red) staining of rosette fusome in E13.5 ovary: GCNA (nuclei, green). (H) Schematic of rosette formation in 4-cell cyst. (I) Plot showing EMA staining loss in germ cells after E13.5. (N=15 per stage). Student's t-test was used for each graph in Figure 1 (*** = $p < 0.001$). Scale bars: 5 μm (A, F, F'), 10 μm (B-B', C-C', D', G-G'), 20 μm (D), 2 μm (E).

The mouse fusome overlaps centrosomes and associates with microtubules

A cell's microtubule cytoskeleton emanates from its centrosomes and from non-centrosomal microtubule organizing centers (ncMTOCs); it frequently controls trafficking of Golgi elements and positions them near centrosomes. To investigate whether the fusome associated with microtubules, we stained E11.5 gonads with anti-acetylated tubulin (AcTub) and EMA. The fusome overlapped the centrosomes in interphase germ cells (Figure 2A [↗](#)), indicating it usually occupies a peri-centriolar domain. In germ cells undergoing mitosis (Figure 2B [↗](#)), the fusome associated with centrioles during interphase, but (typical for Golgi) dispersed as the mitotic spindle forms and for most of metaphase. In telophase, fusome vesicles gather at the spindle poles in both daughter cells and at the arrested cytokinesis furrow, which still retains the elongated spindle/midbody microtubules typical of cyst-forming cell cycles (Figure 2B [↗](#)). Small EMA-positive vesicles often associated with these telophase spindles, providing evidence that fusome vesicles can move along microtubules (Figure 2C [↗](#), Figure S2E [↗](#)). 3D summary movies of germ cells at each mitotic stage stained for AcTub and EMA are shown in Figure S2A [↗](#).

The mouse fusome was rarely seen at just one pole of a mitotic spindle, like the *Drosophila* fusome at early stages of GSC and cystoblast division (de Cuevas and Spradling, 1998 [↗](#)). Some EMA-reactive material appears at the cytokinesis furrow before it is seen at the fusome (Figure 2C [↗](#); Figure S2C-D [↗](#)). Asymmetry in the mouse fusome distribution arose only in late telophase. Clusters of fusome and centrosomes approached the arrested furrow (Figure 2D [↗](#), S2C-D [↗](#)). At this point short microtubule arrays were visible (Figure S2C [↗](#)). Small fusome-containing vesicles are often observed in association with microtubules as shown in Figure S2E [↗](#). Soon after, the amount of fusome in the two daughter cells becomes asymmetric, suggesting that differential activity or persistence of fusome transport late in cytokinesis generates asymmetry. Ovaries cultured briefly in vitro support continued germ cell division and fusome asymmetry. Fusome formation is severely affected in such cultures if microtubules (MTs) are disrupted by cold or by Ciliobrevin D treatment (Figure S2F graph [↗](#)).

Cyst fragmentation is non-random and correlates with microtubule gaps

Mouse cysts produce an average of 4-6 oocytes per PGC as a result of initial cyst breakage into multiple, independently developing subcysts that each produce one oocyte (Lei and Spradling, 2013 [↗](#); Niu and Spradling, 2022 [↗](#)). Production of uniform-sized oocytes, as observed, would seem to require programmed breakage into uniform subcysts (Spradling et al. 2022 [↗](#)), but some random breakage also occurs (Lei and Spradling, 2013 [↗](#); Ikami et al. 2023 [↗](#); Levy et al. 2024 [↗](#)). To assess a role of microtubules and the fusome in programming cyst fragmentation, we tracked AcTub and the fusome in lineage-labeled E12.5 cysts. In about 10% of interphase cysts, presumably those in very late telophase or early in the subsequent cell cycle, persistent microtubule arrays were observed that connected the cells in pairs (Figure 2E [↗](#)). When we examined a total of about 50 large cysts, 15 larger cysts of 7-10 cells were found where the MTs in all cells could be analyzed completely. Three examples are shown in which projections of 8-cell lineage-labeled cysts are outlined (dashed lines). In the left column, lineage, the fusome and MTs are shown (Fig. 2E-E' [↗](#); Figure S2H [↗](#)). Unexpectedly, the microtubule connections (middle column) showed a large discontinuity of 10μ or more between just two of the cystocytes (Figure 2E [↗](#)). Likewise, microtubule bundles in E' and E'' are also absent between two cells (straight dashed line Figure 2E'-E'' [↗](#) and Figure S2H' [↗](#)). The absence of an MT connection predicts cyst breakage, as observed in one of the cysts (Figure 2E'' [↗](#)), where separation into a 2-cell and 6-cell cyst has already occurred. We also examined how the fusome distributes in these cysts (Figure S2H' [↗](#)).

The patterns of cyst breakage predicted by MT absence were highly non-random (Figure 2F [↗](#)). A 6-cell cyst is predicted based on the position of the microtubule gaps in 13/15 8-cell to 10-cell cysts analyzed: see Figure 2E [↗](#), Figure S2H [↗](#) (movies of an 8-cell (upper), or 10-cell cyst (lower), with cleavage diagram); and Figure S2H' [↗](#) (fusome separation in the 3 cysts from Fig. E-E''). A binomial

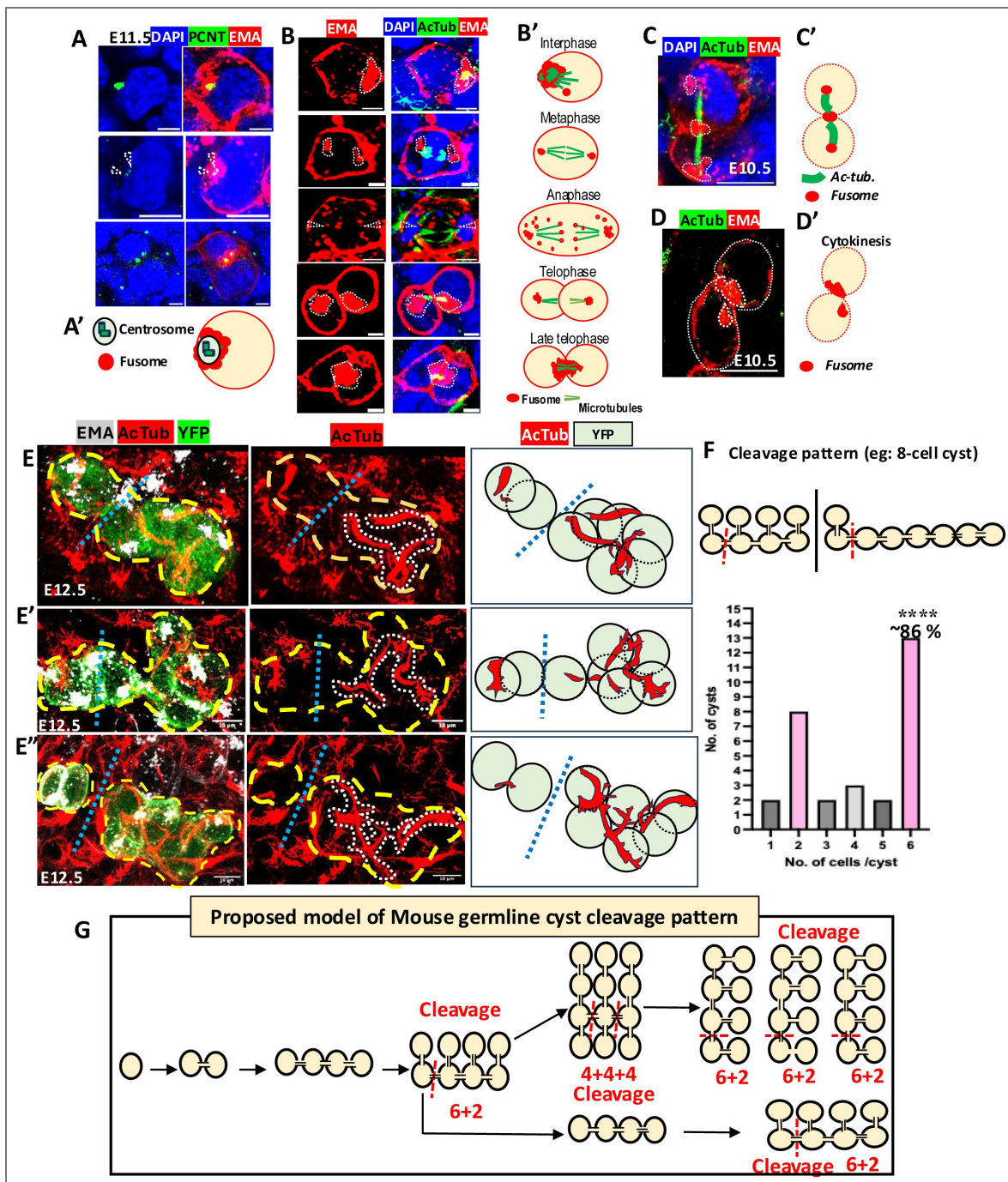


Figure 2. Stabilized spindle microtubules mediate fusome asymmetry and cyst breakage.

A. Pericentric fusome localization in E11.5 germ cells. The early fusome (EMA granule) associates with centrosomes (dashed arrowheads) in E11.5 germ cells: EMA (red), centrosomes PCNT (Pericentrin, green). Left column shows PCNT and DAPI alone. A'. Summary. B. Fusome (EMA) behavior during indicated stages of the 1st cyst cell cycle. B'. Diagrams summarize behavior at the listed mitotic stages deduced by AcTub staining. C, C'. Symmetric early telophase fusome. D, D'. Asymmetric fusome segregation during late cytokinesis. B-D. EMA, acetylated tubulin (Ac. Tub) and DAPI. E-E''. Three lineage-marked (YFP) E12.5 8-cell cysts (E-E'') in early interphase stained to reveal microtubules (AcTub) and fusome (EMA). The absence of spindle remnants in one cyst region (gap in AcTub) predicts future cyst breakage (blue dashed line, summary at R only of YFP+ cells). The E'' cyst has already broken into 2-cell and 6-cell cyst derivatives. F. Frequency distribution predicted cyst breakage products by size based on 15 lineage-labeled cysts analyzed as in E (7-cell: 3; 8-cell: 8; 9-cell:1; 10-cell:3). Binomial test (see text) compared observed 6-cell cyst production frequency (13/15) to prediction for single random junction breakage of an 8-cell cyst). (****p<0.0001). G. Model of cyst production and breakage into four 6-cell cysts and 4 2-cell cysts. Scale bars: 5µm (B), 10µm (A, C-E).

probability calculation showed that the chance of obtaining a 6-cell cyst product in 13 of 15 trials following random breakage of a single junction in an 8-cell cyst (where 2 of 7 broken linkages produce a 6-cell product) was at most 4.53E-06.

The mouse fusome associates with polarity and microtubule genes including *Pard3*

Microtubules play a critical role in ovarian cyst formation and polarity in multiple species. Some of the most important regulators of MTs and polarity are the Par proteins that function in epithelial cell polarity, embryonic development, asymmetric neuroblast division (Petronczki and Knoblich, 2001 [↗](#); Hapak et al. 2018 [↗](#)) and oocyte formation in *C. elegans*, *Drosophila* and vertebrates (Kemphues et al. 1988 [↗](#); Huynh and St Johnston, 2004 [↗](#); Moore and Zernicka-Goetz, 2005 [↗](#)). *Drosophila* meiotic cysts express Par3/Baz and Par6, another apical Par complex protein, along with beta-catenin/Arm in a ring-like arrangement close to the ring canals (Cox et al. 2001 [↗](#); Huynh et al. 2001a [↗](#), 2001b [↗](#); Huynh and St Johnston, 2004 [↗](#)). This relatively small membrane zone likely represents the apical domain of cyst germ cells.

We investigated the expression of *Pard3*, the mouse ortholog of Par3/Baz to determine if a similar arrangement of the highly conserved Par proteins occurs in mouse germline cysts. *Pard3* expression localized around the fusome but extended beyond it, a pattern reminiscent of the larger ring of Baz in *Drosophila* cyst cells (Figure 3A [↗](#), Figure S3A [↗](#)). At E13.5, cysts in rosette configuration show *Pard3* and the fusome enriched around a central cell (Figure 3B [↗](#), dashed region from Video S3B1 [↗](#)) but not in E13.5 male cysts (Video S3B2 [↗](#)). This co-localization was validated in multiple lineage-labeled E13.5 cysts (Figure 3D [↗](#); Figure S3D and S3E [↗](#)). Ring canals were concentrated in this zone (Figure 3C [↗](#) and Figure S3C [↗](#)), hence mouse cysts have ring canals and an apical Par protein in the same small anterior location as in *Drosophila*, and that mouse fusome occupies the corresponding zone as the *Drosophila* fusome. These findings suggest that mammals form and polarize oocytes using a conserved system found in diverse animals based on Par genes, cyst formation and a fusome.

The mouse fusome associates with Golgi genes involved in UPR beginning at the onset of cyst formation

We performed single cell RNA sequencing (scRNA seq) of mouse gonads to gather additional fetal germ cell data (see Methods) and combined them with data from Niu and Spradling (2022) [↗](#) to make an integrated dataset of more than 3,000 germ cells spanning E10.5 to P5 (Figure 3F-H [↗](#); Figure S3F-I [↗](#)). Our gene expression studies showed that PGCs and early cysts actively express many genes that enhance proteome quality by monitoring, refolding or expelling misfolded or aggregated proteins for degradation. The Unfolded Protein Response (UPR) pathway associated largely with the ER in yeast is regulated by Hac1, the ortholog of Xbp1. We observed high levels of Xbp1 (Figure 3I [↗](#)) and many of its target genes (Figure 3I and I' [↗](#)) that increase quality secretory protein production. Membrane proteins and lipids are also promoted by this pathway.

In animals the UPR pathway has been expanded with several additional branches (Mitra & Ryoo, 2019 [↗](#), Sicari, 2019 [↗](#)). We also observed expression Creb3l2 (ER/Golgi resident transcription factor), ATF4 (General UPR transcription factor activated by Golgi stress), Golp3 (Golgi phosphoprotein for membrane trafficking) and Arfgap(s) (involved in both Golgi to ER and Golgi to endosome trafficking) during mouse cyst formation (Figure S3J). Xbp1 activates genes such as the chaperones (Hsp5a/Bip, Hsp90b1; DNAJc10; Lman2; Dnajb9), protein-disulfide isomerases (Erp44, Pdia3, Pdia4, Pdia6, Creld2), the Endoplasmic Reticulum Associated Degradation (ERAD) regulator Syvn1/Hrd1 and the ERAD components (Sec61b; Ubxn4, Psmc6, Ube2j2, Get4, Ufd1) (Taniguchi and Yoshida, 2017 [↗](#)). Thus, repair and expansion of ER, Golgi and secretory components begins immediately or even before migrating PGCs reach the gonad (Figure 3I [↗](#)).

Xbp1 also supports synthesis of ER membrane phospholipids via upregulation of targets such as Elov11, Elov15, and Elov16, and Chka. Other highly expressed targets maintain ER and Golgi structure and are orthologs of *Drosophila* fusome proteins (Figure 3I' [↗](#), Lighthouse et al., 2008 [↗](#)).

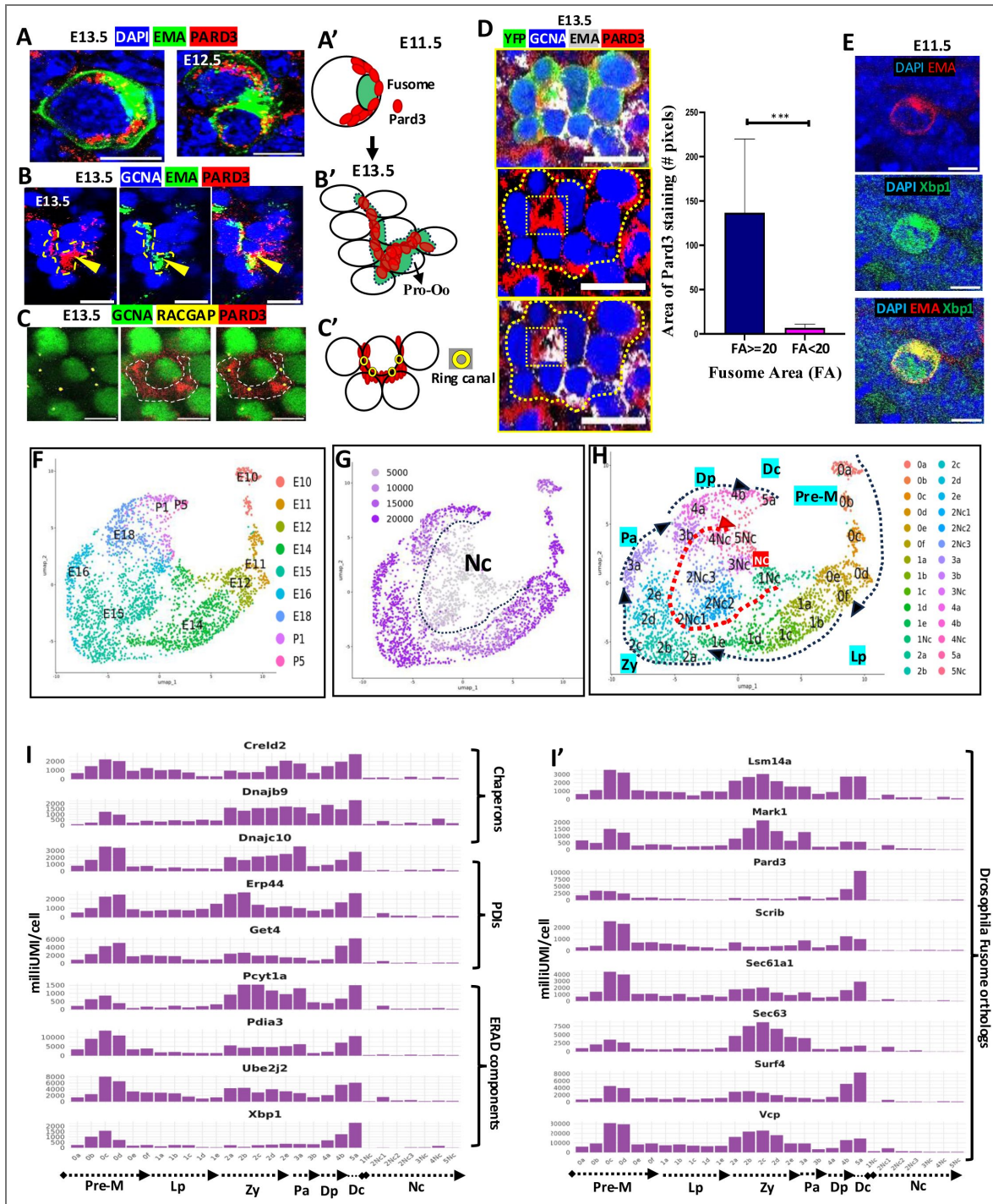


Figure 3. Mouse fusome associates with Pard3 and apical polarity

A-B. Pard3 associates with fusome at E11.5-E13.5 (A, A') and after rosette formation at E13.5 (B, B'). C-C'. Ring canals (RACGAP) localize within the Pard3+ apical domain. D. A lineage-labeled E13.5 cyst (YFP); channels below show enrichment of Pard3 with enriched fusome. Graph: Quantification of Pard3 stained area colocalizing with large $\geq 20 \mu\text{m}^2$ and small $< 20 \mu\text{m}^2$ fusome within lineage labelled cyst (Student's t-test, N=13; ***p<0.001). E. Xbp1 enrichment in EMA granule of E11.5 PGC. F-H. scRNAseq of E10.5-P5 gonad. UMAP of re-clustered germ cells at various stages (F), UMI (G) NC=nurse cells. (H): UMAP with clusters labelled in ascending order of meiotic development. pre-meiotic (Pre-M), leptotene (Lp), zygotene (Zy), pachytene (Pa), diplotene (Dp), dictyate (Dc). (I-I') Bar plots: I. Xbp1, Xbp1-target expression plots. I'. Genes orthologous to fusome components. Scale bars: 10 μm (A-C), 20 μm (D).

Thus the adaptive UPR pathway and its branches are active during cyst formation, and later in prophase as well (Figure 3I–3I').

UPR genes are active during cyst formation and depend on *Dazl*.

We investigated how cyst development and fusome formation depend on *Dazl*. First, we confirmed that *Dazl*^{−/−} mice retard the PGC to germ cell transition (Figure S4C–E). *Dazl*^{−/−} E12.5 cysts showed higher levels of the DNA methylase *Dnmt3a* expression compared to controls (Figure 4A; Figure S4B) suggesting that germ cell DNA demethylation has slowed. Ring canals highlighted by staining for *Tex14* were only half the size of wild type canals at E13.5 or structurally abnormal showing that cyst formation was slowed and cytokinesis was sometimes abnormal (Figure 4B).

We also performed scRNA sequencing of *Dazl* female homozygous mutant gonads at E11.5 and E12.5 and re-clustered them with wild type germ cells at these stages for comparison (Figure 4C). At E11.5 the merged germ cells cluster together suggesting that *Dazl*^{−/−} has minimal effects on germ cells at E11.5, the stage it is first expressed (Figure 4C). However, E12.5 WT and *Dazl*^{−/−} germ cells cluster separately (Figure 4C). Wild type and *Dazl*^{−/−} expressed similar amounts of EGAD mediated UPR pathway genes at E11.5 germ cells (Figure S4F) whereas wild type germ cells expressed lower levels compared to *Dazl*^{−/−} cells at E12.5 (Figure 4C'). Thus, UPR genes in *Dazl*^{−/−} cells act similarly to pluripotency genes, and retain higher expression levels at E12.5 than wild type.

The high expression of *Xbp1* and many of its target genes in PGCs and early cyst cells argues that early germ cells are expanding ER and Golgi production and increasing membrane protein quality. To validate that XBP1 is active in germ cells, we employed an IRE1-XBP1 ratiometric assay which utilizes a genetically encoded dual-fluorescent reporter system (see Methods, Figure S4I). First, we performed magnetic-activated cell sorting (MACS) to purify SSEA1 labelled germ cells from the E11.5 gonad (Figure S4G–H). We delivered the IRE1-XBP1 sensor to SSEA1+ve germ cells and SSEA1-ve control somatic cells by transfection. Both types of cells fluoresce in proportion to IRE1-splicing activity which measures *Xbp1* activation activity. The construct also constitutively expresses a fluorescent marker such that the color ratio will represent a quantitative measure of relative IRE1-XBP1 activity between cells and under different conditions. Our observations showed that IRE1-*Xbp1* activity in E11.5 germ cells was significantly higher than in somatic cells (Figure 4D; Figure S4I). *Dazl* mutant cells showed persistent higher levels of *Xbp1* activity at E12.5 (Figure 4D'–4D'').

The adaptive UPR pathway destroys misfolded proteins using proteasome activity, particularly the 20S proteasome (Vember and Brodsky, 2008; Figure 4H). We measured proteasome activity within E11.5 PGCs to investigate whether the UPR pathway was paired with increased proteasome activity indicating it served an adaptive cellular mechanism of cell rejuvenation in PGCs destined to produce oocytes. Using a fluorogenic substrate to measure 20S Proteasome activity, we compared MACS sorted germ cells to somatic cells and found that germ cells have significantly higher proteasome activity than somatic cells at E11.5 (Figure 4E) and E12.5 (4E' Left). Proteasome activity continued to stay even higher in E12.5 *Dazl*^{−/−} germ cells suggesting stressful persistent activation of UPR keeping *Dazl*^{−/−} germ cell in an arrested state (Figure 4E' Right, 4E'').

We also studied the effects of *Dazl* mutation on the asymmetric enrichment of fusome in cells with multiple ring canals. Using both EMA and a Golgi marker, *Gs28*, we observed that germ cell Golgi begins to fragment in *Dazl*^{−/−} mutant germ cells even by E12.5 (Figure 4F, dotted circles, graph). High levels of Golgi stress documented by the increased IRE1-*Xbp1* activity, and elevated proteolysis in these cells, may lead to such fragmentation. These problems were accompanied by a dramatic arrest in cyst polarity development. *Dazl*^{−/−} germ cells at E13.5 showed virtually no differential expression of *Pard3* or enrichment of fusome within particular cyst cells (Figure 4F'). Finally, in *Dazl*^{−/−} P0 gonads, large cells destined by become oocytes do not appear (Video S5E3, Video S5F3). The effects of *Dazl*^{−/−} on germline cyst development and meiosis are summarized in Figure 4G. These observations collectively indicate an essential function of the

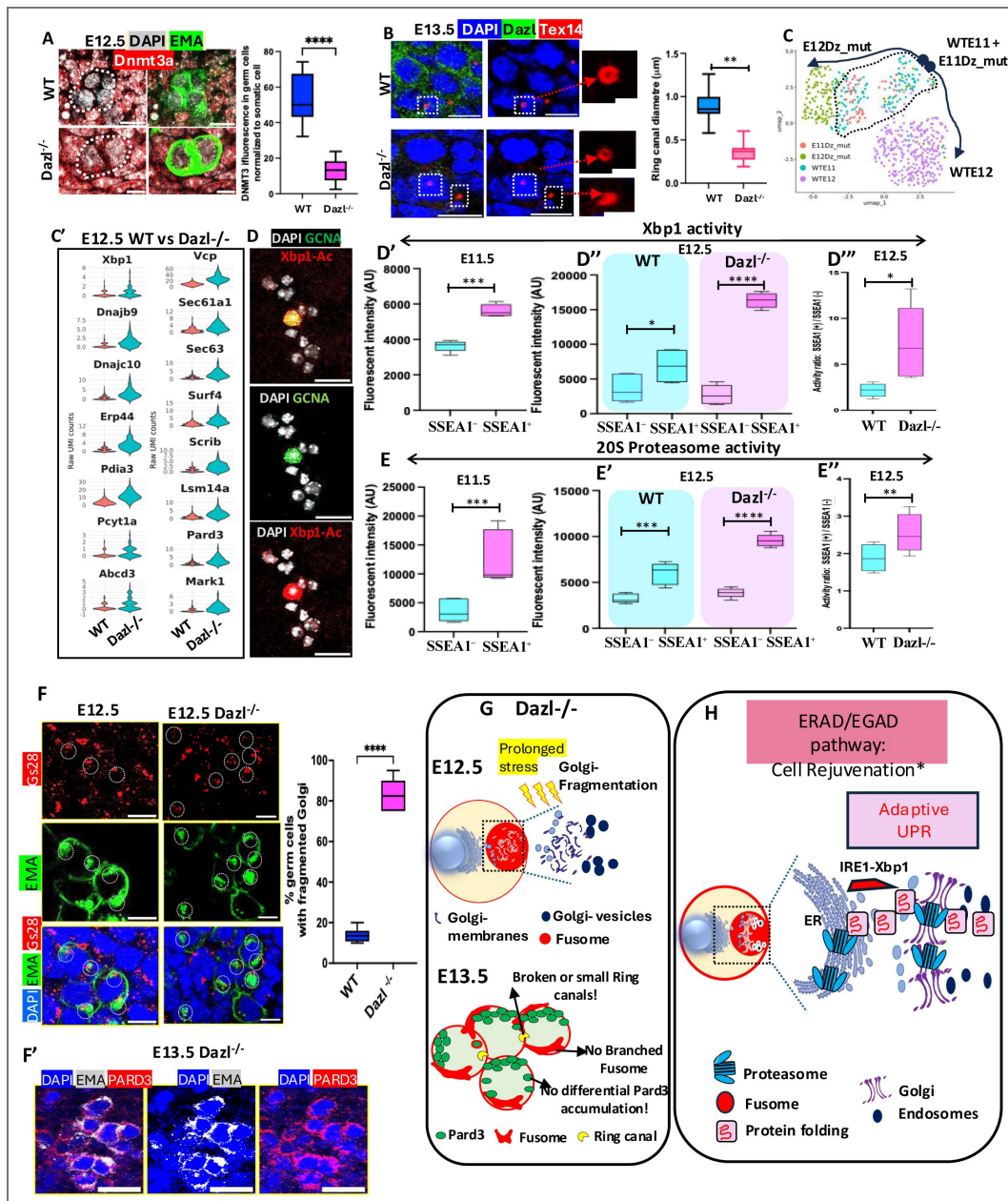


Figure 4. UPR genes are active during cyst formation and controlled by Dazl

(A) Dnmt3a and EMA levels at E12.5. Dnmt3a levels are reduced in WT compared to Dazl^{-/-} germ cells. Graph-Dnmt3a fluorescent levels within germ cells as normalized with somatic cells in WT versus Dazl mutant gonad. (N=10 tissues; **p<0.05). (B) Ring canals are smaller and defective in E13.5 Dazl^{-/-} cysts compared to WT. (N=44; **p<0.05). (C) scRNA seq of E11.5 and E12.5 WT and Dazl^{-/-} gonad germ cells. UMAP. Germ cell clusters overlapped at E11.5 and segregated at E12 of WT and Dazl^{-/-}. (C') Xbp1, Xbp1 targets, and fusome orthologs in WT vs Dazl^{-/-} germ cells. (D) Validation of IRE1-Xbp1 assay: Ovarian cells visualized by fluorescent microscopy showing GCNA labelled bigger germ cells with higher Xbp1 fluorescence than smaller somatic cells (D'-D''). IRE1-Xbp1 assay comparing SSEA1+ germ vs SSEA1- somatic cells at E11.5 and WT vs Dazl^{-/-} germ cells at E12.5. (D'; 6 experiments: ~32 mice, ≥5 mice and ≥20 ovaries per experiment, D''-D'''; 3 experiments: ~40 mice, ≥5 mice and ≥25 ovaries per experiments, *p<0.05, **p<0.01, ***p<0.005, ****p<0.0001) (E-E') Proteasome activity comparing SSEA1+ germ vs SSEA1- somatic cells at E11.5 and WT vs Dazl^{-/-} germ cells at E12.5. (N=3 biological assays with ~35-60 E11.5 ovary per assay and ~25-28 E12.5 ovaries were used per assay. *p<0.05, **p<0.01, ***p<0.005, ****p<0.0001) (F) Golgi fragmentation in E12.5 Dazl^{-/-} germ cells. Graph: germ cell percent with fragmented Golgi in wildtype versus Dazl mutant mouse gonad (Student's t-test: N=16, ***p<0.005) (F') Failure of E13.5 Dazl^{-/-} germ cells to form EMA rosettes or enrich Pard3. G. Dazl^{-/-} effects on fusome, Golgi and Pard3. H. Proposed function of fusome mediated regulation of ERAD-UPR proteostasis. Scale bar: 10μm (except zoomed in 2μm).

mouse fusome in temporal regulation of the EGAD-UPR pathway within germ cells (Figure 4H [↗](#)). The observed loss of polarity caused by *Dazl*^{-/-} mutation can explain the failure of oocyte production and the sterility of *Dazl* homozygotes.

The mouse fusome is associated with organelle rejuvenation during meiosis

In *Drosophila*, starting at pachytene the microtubule cytoskeleton changes its structure, and polarity (Cox and Spradling, 2006 [↗](#)). The new polarity, with microtubules running along the fusome such that minus ends cluster in the oocyte, prepares selected nurse cell organelles for transport into the oocyte and the Bb of a new primordial follicle. Mouse nurse cell centrosomes, mitochondria, Golgi, ER and possibly other organelles also move from nurse cells to the oocyte between P0 and P4 in a microtubule-dependent process (Lei and Spradling, 2016 [↗](#); Niu and Spradling, 2022 [↗](#)). Prior to or during these events, *Drosophila* mitochondria are selected and rejuvenated by programmed mitophagy to enhance functionality (Cox and Spradling, 2003 [↗](#); Lieber et al. 2019 [↗](#); Palozzi et al. 2022 [↗](#); Monteiro et al. 2023 [↗](#)). It is not currently known if selection and rejuvenation take place before movement to the oocyte, or if a significant amount also takes place en route.

Because the fusome is directly involved in organelle movement to the *Drosophila* Balbiani body we looked for changes in the fusome and in gene expression that might relate to organelle rejuvenation and movement to the oocyte. For the reasons explained previously (see text for Figure 1G [↗](#)) we used WGA as fusome marker beyond stage E14.5. Fusome staining with WGA persists within meiotic germ cells at E17.5 (Figure 5A [↗](#); Figure S5A-B [↗](#)). Fusome volume remained in proportion with germ cell ring canal number, as outer nurse cells (with one ring canal) transfer organelles and cytoplasm to cells with more ring canals located closer to oocytes (Figure 5A, graph [↗](#); Video S5C [↗](#)). *Pard3* content also showed such a proportion (Figure 5A' graph [↗](#)). At E18.5, cyst development in the medulla toward wave 1 follicles, diverged from wave 2 follicles in the cortex that become quiescent primordial follicles (Figure 5B [↗](#)) (Yin and Spradling, 2025 [↗](#)). Wave 1 oocytes grow substantially in volume, fusome and *Pard3* content due to cytoplasmic transfer from surrounding nurse cells which decrease in size (Figure 5B-B', graph [↗](#), Video S5E1 [↗](#), Video S5F1 [↗](#)). Oocyte enrichment of the fusome and *Pard3* was confirmed using lineage labelled cysts at P0 (Figure 5C [↗](#) and 5D [↗](#)). Such enrichment was significantly reduced in *Dazl*^{+/-} heterozygotes (Figure 5E graph [↗](#), Video S5E2 [↗](#), Video S5F2 [↗](#)).

Wave 2 follicles which produce nearly all fertile oocytes, increase organelle content beginning at E18.5 and contain a finished Bb by P4 (Niu and Spradling, 2022 [↗](#)). Initially, 4-5 transferred Golgi-centrosome pairs were observed at separate cytoplasmic locations in oocytes. These gradually come together and form an oval-shaped cytocentrum that nucleates a microtubule aster and organizes the Balbiani body. We carried out immunofluorescent staining at E18.5 and examined EMs that documented interactions between Golgi, ER and mitochondria. Figures 5F-F' [↗](#) show a wild type germ cell with a large crescent of perinuclear mitochondria (AT5a) adjacent to a large cluster of fusome (WGA). An EM provides a higher resolution view of a similar interaction (Figure 5F' [↗](#)). In contrast, relatively few mitochondria or fusome were stained in a *Dazl*^{+/-} heterozygote (Figure 5F [↗](#)). Mitochondria also interact extensively with ER at this time (Figure 5G [↗](#); Pepling et al. 2007 [↗](#)). ER tracker and Mitotracker (Figure 5G' [↗](#)) stained clusters throughout much of the germ cell cytoplasm.

Discussion

Mouse germline cysts contain a fusome

One of the earliest and most highly conserved steps in animal gamete development is the formation of polarized germline cysts (Giardina, 1901 [↗](#); Wilson, 1925 [↗](#); Büning, 1994 [↗](#); Matova and Cooley, 2001 [↗](#); Lu et al. 2017 [↗](#); Gerhold et al. 2022 [↗](#); Spradling et al. 2022). In the ovaries of most species, cysts become polarized during the synchronous mitotic cycles whose incomplete cytokinesis connects daughters via characteristic ring canals. In *Drosophila*, germline cyst

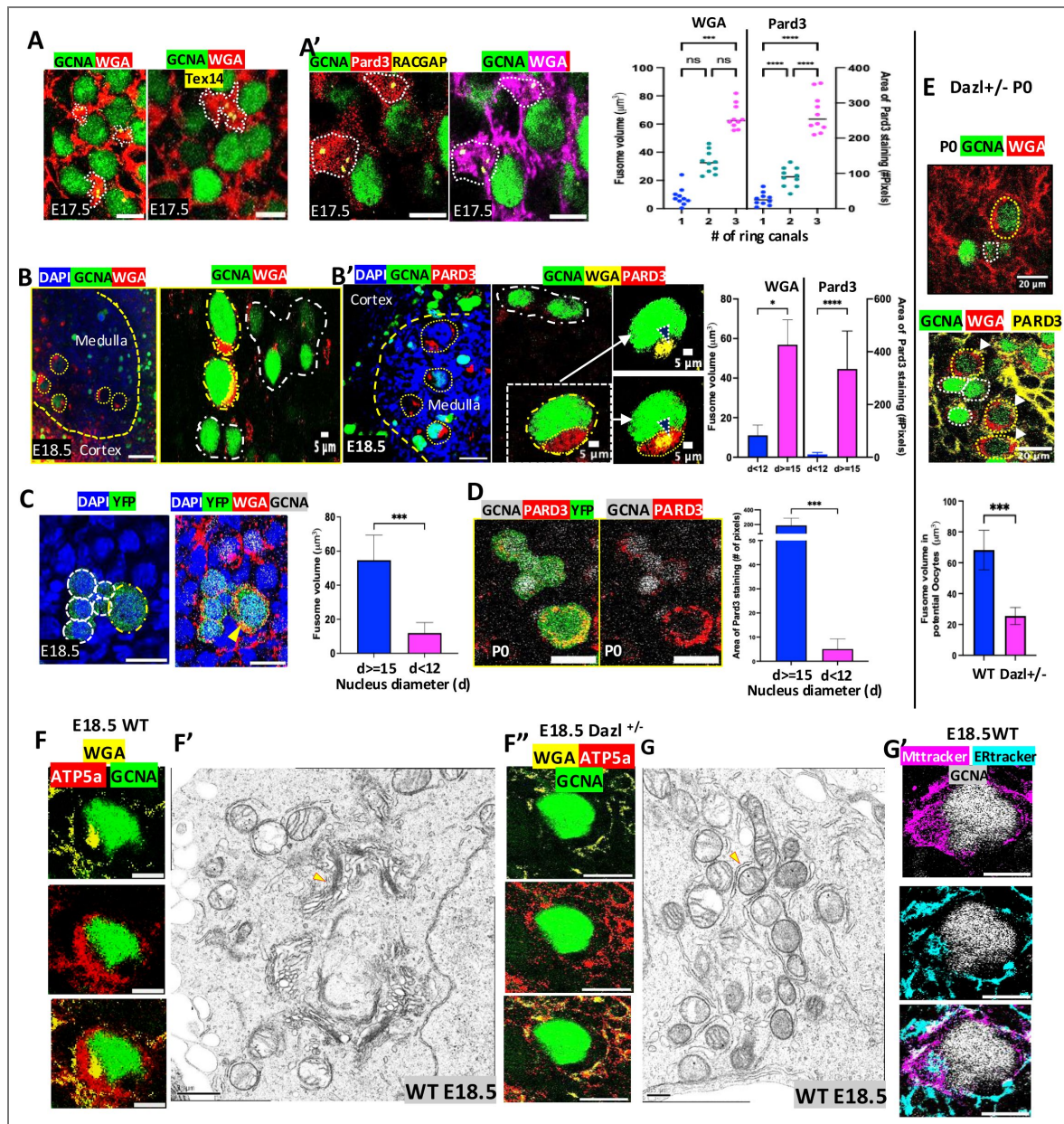


Figure 5. Fusome and Pard3 Associate with ER and mitochondria prior to Balbiani body formation

A. E17.5 ovary stained for WGA, GCNA, and TEX14. A'. E17.5 ovary stained for GCNA, RACGAP and PARD3; Graph: Fusome volume and Pard3 Stained area versus ring canal number (N=65 (Fusome volume); N=51 (Pard3); ANOVA, *** $p < 0.005$, **** $p < 0.0001$). B-B'. E18.5 ovary shows WGA-Fusome/Pard3 enrichment in large medullary oocytes vs smaller nurse cells; line: medulla/cortex boundary; dotted circle: large medullary oocytes; white dotted area: small nurse cells. Area marked as white dotted rectangle is shown as zoomed inset (white arrow). Black arrow in inset: WGA stained fusome; Graph compares fusome volume and Pard3 stained area versus Germ cell nucleus diameter (N=54 (WGA), N=37 (PARD3); Student's paired t-test, * $p < 0.05$, **** $p < 0.001$). C. Single cell lineage labelled E18.5 ovary stained for YFP, DAPI, WGA and GCNA Graph: Within single-cell lineage-labeled E18.5 ovary-Fusome volume difference according to germ cell nucleus size (N=10; **** $p < 0.0001$). D. Single cell lineage labelled E18.5 ovary stained for YFP, PARD3 and GCNA Graph: Within single-cell lineage-labeled E18.5 ovary-difference in PARD3 stained area according to germ cell nucleus size (N=10; *** $p < 0.005$). E-E'. Dazl^{+/-} E18.5 ovary-Fusome (WGA) and Pard3 enrichment failure in medullary Oocytes (GCNA). Graph: Fusome volume in potential Oocytes i.e., bigger germ cells with nucleus diameter $d \geq 12 \mu\text{m}$ in Wildtype versus Dazl^{+/-} mutant F-F'. Organelle enrichment analysis: E18.5 (WT-F-F', and Dazl^{+/-} ovary-F'') stained for WGA, mitochondrial marker ATP5a and GCNA (F and F''). F'-EM image of Golgi-rich Fusome (arrow) surrounded by mitochondria. G-G'. ER-mitochondria association in E18.5 WT ovary: G-EM image of ER tubules (arrow) wrapping mitochondria and G'- E18.5 WT ovary-GCNA, ER and Mitochondria tracker staining. Scale bars: 20 μm (A-E, G-G', F, F'', G'), 5 μm (B, B'-right most inset panel), 0.5 μm (EM images F', G).

divisions also produce and asymmetrically distribute a microtubule and vesicle-rich fusome whose polarity is essential to specify oocytes in addition to nurse cells. We found that mouse germline cysts also contain a similar organelle, the “mouse fusome.” As in *Drosophila*, the mouse fusome starts as a small, membrane-rich cytoplasmic region within migrating PGCs before they arrive at the gonad. The nascent organelle propagates asymmetrically during synchronous, incomplete cyst divisions, involving stabilized spindle microtubules rich in AcTub and a program of arrested cytokinesis that produces distinctive ring canals. Stable fusome MTs persist for only part of each cell cycle, but polarity is maintained since over five successive mitotic cycles the cyst undergoes rosette formation, predicts cleavage into smaller cysts and concentrates fusome material in future oocytes. Finally, the mouse fusome associates in a distinctive manner with the apical Par complex located near the clustered ring canals, as originally described for the fusome in *Drosophila* cysts (Cox et al. 2001 [DOI](#); Huynh et al. 2001a [DOI](#), 2001b [DOI](#)). Apical-basal patterning of *Drosophila* oocytes differs from many epithelial cells (review Huynh and St Johnston, 2004 [DOI](#)), but our studies show that both mouse and *Drosophila* oocytes localize the fusome and ring canals in a similar relationship to the apical Par domain.

How conserved are fusomes in distant animal groups?

The recent detailed analyses of fusomes in *Xenopus* (Davidian and Spradling, 2025 [DOI](#)) and here in mice, provide an opportunity to compare fusomes over a wider range of animals than was possible when well characterized fusomes were limited to Holometabolous insects. Based on these examples, the most highly conserved aspects of polarized cyst production are the cell cycle changes that arrest telophase, stabilize the midbody and lead to incomplete cytokinesis (Gerhold et al. 2022 [DOI](#); Price et al. 2023 [DOI](#)). Somehow this process polarizes the microtubules that arise in the fusome segment within the nascent ring canal, and becomes a building block in growing cyst polarity. A longstanding mystery regarding fusome microtubule polarity is its exceptionally high sensitivity to microtubule inhibitors like colchicine, which appears to be much greater than cell division generally, or other developmental processes requiring microtubule-dependent vesicle transport (Koch and Spitzer, 1983 [DOI](#)). The sensitivity continues later in meiosis, since treating E17.5 mouse ovary cultures with just 10 nM colchicine strongly reduced Balbiani body formation, but had no measurable effect on cell division (Lei and Spradling, 2016 [DOI](#)). Interestingly, mouse cysts develop asymmetry during a later period in cytokinesis than *Drosophila* cysts (Figure 2C–D [DOI](#)). In *Xenopus*, another fusome-related property, cyst-wide divisional synchrony, showed more variation between sister cells than in tightly coordinated *Drosophila* cyst cells (Davidian and Spradling, 2025 [DOI](#)).

Qualitative differences in fusome structure and behavior have evolved between major animal groups. *Drosophila* fusomes and those of related insects are stabilized by an essential, proteinaceous skeleton containing alpha-spectrin, tropomodulin and germline-specific isoforms of adducin-like proteins generated by differential splicing and cleavage of a polyprotein precursor produced by the Hts gene (Petrella et al. 2007 [DOI](#)). This skeleton may have evolved to control access to the fusome and hence to the oocyte by intracellular parasites such as *Wolbachia* bacteria. An alpha-spectrin-containing skeleton was not detected in *Xenopus* fusomes (Davidian and Spradling, 2025 [DOI](#)) or in these experiments in mouse. The *Xenopus* fusome has robust microtubules, that may independently control access. This structural difference may explain earlier failures to identify fusomes in vertebrates using orthologs to *Drosophila* fusome proteins like alpha-spectrin.

Association of the mouse fusome with Golgi

The major role apparently played by Golgi in the mouse compared to the *Drosophila* or *Xenopus* fusomes is potentially significant. In contrast, the *Drosophila* fusome contains abundant ER cisternae, whose presence may facilitate lipid biosynthesis needed to produce high levels of new plasma and organellar membranes. Not does rapid cyst formation demand high amounts of fresh membranes, but rejuvenation of new germ cells may require that their lipids and proteins be newly synthesized from simple building blocks (see Spradling et al. 2025 [DOI](#)). Mouse cysts grow more slowly, and may associate extensively with Golgi because of an early high demand for

protein secretion. In mouse, the zona pellucida directly contacts the oocyte and begins to be produced and secreted by the oocyte in primary follicle oocytes have just begun to develop. In contrast, the analogous *Drosophila* egg coating, the vitelline membrane, is produced by somatic follicle cells, and only starting in more mature follicles. Even primordial follicles oocytes contain abundant Golgi and multivesicular bodies near the Balbiani body (Niu and Spradling, 2022). Thus, fusome's apparently stronger association with the Golgi in mouse but with the ER in *Drosophila* probably just reflects differences the relative relative timing and speed of events in oogenesis that take place in both organisms. This highlights the need to study fusome behavior throughout oogenesis and in the context of ongoing developmental and rejuvenation activity.

Mouse cysts develop multiple uniform oocytes using a novel specific-cleavage mechanism

Mouse PGCs generate 4-6 oocytes using a multi-step mechanism involving synchronous incomplete cyst divisions, limited fragmentation into smaller cysts (involving <10% of cell-cell junctions) and continuous, slow nurse cell turnover (Lei and Spradling, 2013; Lei and Spradling, 2016; Niu and Spradling, 2022). Despite their complex pathway, mouse oocytes in newly formed primordial follicles appear quite uniform in size and are efficiently produced (Spradling, 2022), suggesting that a majority develop from final cysts of similar size and nurse cell content. However, lineage analyses showed many cyst cells move apart at least transiently. Live imaging studies revealed active movements likely responsible for cell displacement, as well as some cell detachment caused by random breakage (Levy et al. 2024).

Our studies of mouse fusome microtubules indicated that oocyte uniformity is promoted by programmed cyst breakage during mitotic divisions into cysts of uniform size (Figure 2G). ActTub arrays in cysts of 7-10 cells frequently contained a single large gap lacking MTs between a group of 6-cells, and the other residual cells. Such gaps in a major cytoskeletal element of germ cells subjected to significant tissue movements would likely become breakage sites (Figure 2E''). Previous experiments also supported a special role for 6-cell cysts. We conclude that germline cyst breakage is largely programmed and efficient 6-cell cyst production explains the high frequency of synchronous six-cell mitoses observed during E11.5-E13.5 (Pepling and Spradling, 1998) and directly mapped 6-cell cysts in young cyst stages prior to E14.5 (Lei and Spradling, 2013). Efficient cell turnover and transfer of most nurse cell contents into the oocyte probably explains the 5.1-fold increase in Golgi, 4.9-fold increase in mitochondria and 5-fold increase in the number of accumulated centrosomes measured in P4 oocytes compared to E14.5 germ cells (Lei and Spradling, 2016).

How might growing cysts be programmed to acquire a gap in remnant spindle microtubule bundles affecting a specific junction? One possibility is the uneven distribution of the fusome in four cell cysts (Figure 1B''). If the fusome is involved in stabilizing and associating with spindle remnants, shortly after M phase, the cell with the lowest fusome amount might fail acquire a spindle remnant of the necessary size at the next division, leading to 6:2 breakage. Involvement of the microtubule cytoskeleton in asymmetric divisions has been observed in many other situations (Fichelson & Huynh, 2007; Kaltschmidt & Brand, 2002; Sunchu & Cabernard, 2020; Meiring et al., 2020; Planelles-Herrero et al., 2022; Watson et al., 2023). Significantly, other sources of variation likely also occur at a lower level. Synchronously dividing 8-cell and 16-cell cysts are readily detected in low numbers (Pepling and Spradling, 1998). These large cysts may represent a specific alternative pathway, that may preferentially take place in cysts within one of the three follicle waves that occur in mouse (Niu and Spradling, 2025). The model in Figure 2G, also predicts production of 4 2-cell cysts which were commonly observe in prior to E14.5 and the onset of substantial nurse cell transfer (Lei and Spradling, 2013, Supplement).

Dazl regulates cyst formation, fusome polarization, and oocyte specification

Female gamete development in many organisms is extensively controlled at the post-transcriptional level by RNA binding proteins (Mercer et al 2021 [↗](#); Conti and Kunitomi, 2024 [↗](#)). In vertebrates, the RNA-binding protein Dazl is widely conserved and controls multiple aspects of early germ cell development by regulating a large number of target transcripts (Zagore et al. 2018 [↗](#)). In mice, Dazl modulates the ongoing reprogramming of new arrived gonadal PGCs to pluripotency beginning at E11.5 when it turns on (Gill et al. 2011 [↗](#); Nicholls et al. 2019 [↗](#); Haston et al. 2009 [↗](#)). Normally, Dazl continues to function at later stages of oocyte development (Conti and Kunitomi, 2024 [↗](#)), but Dazl mutants fail to enter or progress in meiosis.

Our studies defined a series of new functions for Dazl during cyst formation. We confirmed that pluripotency reactivation is slowed in Dazl mutants since DNA demethylation was delayed, and pluripotency genes failed to start downregulating at E12.5. Homozygous Dazl mutant females failed to normally regulate UPR genes, leaving their mRNA levels elevated. This likely resulted from a failure to downregulate their Xbp1 activator, since both Xbp1 activity and 20S proteasome activity was also elevated in Dazl^{-/-} germ cells. Overactivation of the UPR pathway appeared to cause Golgi fragmentation. These effects resembled the general slowing of normal developmental progression, as with pluripotency downregulation. We also observed abnormal cyst formation with some small and defective ring canals as in zebrafish Dazl mutants (Bertho et al. 2021 [↗](#)), but cysts remained intact and continued to develop. This allowed us to document that Dazl is needed for cysts and their fusomes to become polarized. In Dazl mutants, cysts did not undergo rosette formation or concentrate the fusome or Pard3 in cells with multiple ring canals. Oocytes apparently could not be specified and did not form. These results show that Dazl acts as a regulator that is essential for the many important aspects of cyst formation and oocyte production.

Germ cell rejuvenation is highly active during cyst formation

Germ cells have long been known to propagate species without know limits over evolutionary time by rejuvenating gametes each generation in association with meiosis (Weissmann, 1892; Kirkwood, 1987 [↗](#)). Recently, the detailed cellular and molecular genetic mechanisms that make this possible both in single-celled eukaryotes and in animals have generated increasing interest (Cox and Spradling, 2003 [↗](#); Unal et al., 2011 [↗](#); Chen et al. 2014 [↗](#); Bohnert and Kenyon, 2017 [↗](#); Lieber et al., 2019 [↗](#); Palozzi et al., 2022 [↗](#); Spradling et al., 2022 [↗](#); Yamashita, 2023 [↗](#); Xiao and Unal, 2025 [↗](#)). By studying the fusome, which is already present in migrating PGCs and E9.5, and by sequencing mouse germ cells from the earliest stages of ovarian development beginning at E10.5, we gained new insights into the timing and processes that likely contribute to germ cell rejuvenation during mouse oogenesis.

These studies made it clear that multiple rejuvenation mechanisms are highly active from the earliest times of germ cell development (Figure 3I [↗](#)). In animals the UPR pathway has been modified by the addition of several additional pathways besides the Ire1-Xba1 branch, all of which were expressed in early germ cells. Creb3l2 controls a transcription factor that increases production COPII vesicles that carry out ER to Golgi transport and maintain a balance between secretory supply and demand. Insufficient matching can lead to Golgi stress (Machemer, 2015), which can be managed by activating the ATF4 pathway branch.

Expression of Xbp1mRNA or even Xbp1 protein using a general antibody is not sufficient to show Xbp1 activity, since this is controlled by Ire1-mediated splicing to produce the active Xbp1s isoform. We were able to purify enough E11.5 and E12.5 germ cells to carry out biochemical assays and show that mouse female germ cells have substantial levels of Xbp1s activity at both times, higher than in somatic cells. The high expression of many known Xbp1 targets also documents the high activity of this pathway (Figure 3I,I' [↗](#)). Early germ cells also express genes suggesting that they are expanding membranes via lipid biosynthesis and secretory pathway activity. Consistent with active adaptive-UPR rejuvenation of their proteomes we documented substantial levels of active proteolysis in early germ cells at both E11.5 and E12.5 The observations reported here

suggest that EGAD—an emerging proteostatic mechanism plays a previously unappreciated role. EGAD complements the well-established ERAD pathway by facilitating the clearance of misfolded proteins via Golgi-endosome trafficking and proteolysis, rather than removal from the ER and degradation in the proteasome as in ERAD. We observed substantial amounts of ERAD based on the expression of Xbp1 and its many targets, and its documented activity and its endpoint 20S proteasome activity. It is hard to quantitatively compare the amount of EGAD activity, but the prominent role Golgi throughout the stages of mouse oogenesis and its location at the core of the Balbiani body suggests that EGAD either fulfills a unique role in oocyte production, or else is uniquely capable of operating at the highest level of proteostatic activity demanded by the much greater challenge of germline rejuvenation compared to any other tissue without generating damaging and self-limiting side effects.

Rejuvenation genes were also highly expressed later in meiosis during zygotene and pachytene, in diplotene when mitochondria/ER contact is extremely high (Figure 5G [↗](#)), as well as in primordial follicles (Figure 3I,I' [↗](#) “Dc”). The early appearance of the fusome and its association with Golgi and ER-related processes that are central to adaptive-UPR, suggests that in animals, rejuvenation begins earlier in the generational cycle than it does in single celled eukaryotes. Animal germ cells return to a pluripotent state by the end of cyst formation. Cyst formation precede meiosis and represents an opportunity to enhance the rejuvenation process (Spradling et al. 2025 [↗](#)). Our findings of early rejuvenation activity align with those of Palozzi et al. (2022) [↗](#), who showed that mitochondrial rejuvenation during oogenesis uses a special germline mitophagy and begins at the onset of meiosis. In yeast, mouse and *C. elegans* oocytes, damaged materials are destroyed by lysosome-like degradation in mature oocytes and at meiosis II (Bohnert and Kenyon, 2017 [↗](#); Zaffagnini et al. 2024 [↗](#); Xiao and Unal, 2025 [↗](#)), similar to the lysosome-like turnover of remnant nurse cells, and residual fusome contents near the onset of follicle formation in mouse and *Drosophila*. Thus, animal oocytes may engage in rejuvenation activities from before meiosis begins until meiosis II is completed.

In sum, our experiments argue that mouse germline cysts are produced by an ancient conserved process that modulates spindle microtubules to generate an asymmetric polarized fusome associated with apical Par proteins that enrich in oocytes. Proteins and processes involved in cellular rejuvenation are highly active throughout meiotic prophase and begin even in migrating PGCs. Germline cyst polarity ensures that organelles gathered from the nurse cells are delivered to the oocyte where they form a Balbiani body. In *Drosophila*, germline cysts, fusomes and Par-mediated polarity are all essential for oocyte production. The parallels shown here across 500 million years of evolution strengthen the case that germline cysts and polarized fusomes are likely to have been maintained in evolution because they contribute to the existential task of rejuvenating germ cells each generation,

STAR* Methods

Reagent or Resource	Source	Identifier
Antibody/Lectin/Stains		
EMA	DSHB	EMA-1 AB_531885
GCNA	Abcam	ab82527
DDX4	Abcam/R&D systems	ab13840 & ab27591/ AF2030
SSEA1	DSHB	MC-480
WGA	Thermo Fisher	W7024
Gm-130	ThermoFisher/Novus Biologicals	610822/NBP2-53420
Rab9	Thermo Fisher	MA5-31997
Rab11a1	Thermo Fisher	700184
Tex14	Proteintech	18351-AP
GFP/YFP	Aves Labs	GFP-1020
Dazl	BioRad Labs/GeneTex	MCA2336/GTX89448
Lysotracker deep	Thermo Fisher	L7528
Pard3	Novus Biologicals	NBP1-88861
LAMP1	Cell Signaling Technology	9091T
Xbp1	Abcam	ab37152
Gs28	BD biosciences	61184
AAL	Vector Labs Inc	FL-1391-1
LCA	Vector Labs Inc	FL-1041-5
Sec63	Thermo Fisher	PA5-100180
Calnexin	Abcam	ab219644
Gs28	Proteintech	16106-1-AP
Acetyl- α -Tubulin (Lys40)	Cell Signaling. Tech	5335S
Alpha Tubulin (acetyl K40)	Abcam	ab289875
Pericentrin	Abcam	ab4448/ab28144
Anti-GCNA1 antibody	Abcam	ab82527
Rac GAP1 Antibody (A-6)	Santa Cruz Biotechnology, Inc	sc-271110
Dnmt3a	Cell Signaling. Tech	3598T
ATP5A	Abcam	ab14748
Mitotracker /Deep	Thermo Fischer	M7514/ M22426
ER-Tracker™ /	Thermo Fischer	E34251/ E34250
Quinacrine dihydrochloride	Sigma chemical	Q3251
Chemicals		

Key resources table.

Paraformaldehyde	Electron Microscopy Sci.	15714
Tamoxifen	Sigma Chemical	T5648
Corn Oil	Sigma Chemical	C8267
Tween 20	Sigma Chemical	P1379
Triton X	Sigma Chemical	X100
Mounting media	Vector Labs Inc.	H-1000
Trypsin-EDTA (0.25%)	Fisher	25200056
Fetal Bovine serum	Sigma Chemical	F2442
Bovine serum albumin	Sigma Chemical	A4503
SSEA1 (CD15) microbeads	Miltenyi Biotech	130-094-530
DMEM/F-12	Fischer	11320-033
Celiobrevin D	Millipore	250401

Critical Commercial kits

Kappa fast hot start ready-mix	KAPPA Biosystems	KK5608
KAPPA express extract	KAPPA Biosystems	KR0383-v4.16
10X Genomics kit		
IRE1-Xbp1 Assay	Montana Molecular	U0921G
Proteasome Activity	Amplite	13456
Superscript IV cells direct cDNA synthesis kit	Fischer	11750510

Experimental models

C57BL/6J	Jackson Laboratory	000664
CAG-Cre-ER	Jackson Laboratory	004682
R26R-EYFP	Jackson Laboratory	006148
Dazl1L ^{-/+}	Jackson Laboratory	035880

Others

MACS MS separation columns	Miltenyi Biotech	130-042-201
Cell culture inserts	Millipore	PICM01250
Cell strainer	Sigma	CLS431752

Software and Algorithms

FIJI	FIJI	Version: 2.16.0/1.54p
IMARIS	Oxford Instruments (formerly Bitplane)	Version: 10.2
SEURAT	Satija Lab	Version: 5.1.0
CELL RANGER PIPELINE	10X genomics	Version: 6.0.1/8.0.1
GRAPHPAD PRISM	GraphPad Software	Version: 10.5.0

Key resources table. (continued)

Primer Name	Sequence	Amplicon Size and Reference
Uba1_Foward	5'-TGGTCTGGACCCAAACGCTGTCCACA-3'	Male- 198 bp Female- 217 bp
Uba1_Reverse	5'-GGCAGCAGCCATCACATAATCCAGATG-3'	McFarlane et al., 2013 (https://doi.org/10.1159/000348677)
Sly_Foward	5'-GATGATTGTGAGTGGAATGTGAGGTA-3'	Male- 280 bp Female- 685 and 480 bp
Sly_Reverse	5'-CTTATGTTTATAGGCATGCACCATGTA-3'	McFarlane et al., 2013 (https://doi.org/10.1159/000348677)
Zfy_Foward	5'-GACTAGACATGTCTTAACATCTGTCC-3'	Male- 180 bp
Zfy_Reverse	5'-CCTATTGCATGGACTGCAGCTTATG-3'	McFarlane et al., 2013 (https://doi.org/10.1159/000348677)
DDX4 Forward	5'-GAGATTGCCTTCAGTACCTATGTG-3'	105 bp
DDX4 Reverse	5'-GTGCTTGCCCTGGTAATTCT-3'	Gao et al., 2011 (https://doi.org/10.1095/biolreprod.110.084590)
GAPDH Forward	5'-GGTGAAGCAGGCATCTGAGGG-3'	246 bp
GAPDH Reverse	5'-GGTGGGTGGTCCAGGGTT-3'	Chunmei et al., 2011 (https://doi.org/10.1093/glycob/cwq152)
Dazl Common	5'-GAC ATT ACT AAG AAA ACA GCA GTG G-3'	Mutant=300 bp, Heterozygote=300 bp and 192 bp, Wild type=192 bp
Dazl WT reverse	5'-TTC TGC ACA TCC ACG TCA TT-3'	JAX Protocol (Strain #035880, Primer# 56340, 56341 and 56342)
Dazl Mut Reverse	5'-ATC CCT CCC TTT AGG GCT CA-3'	

Primer details.

Resource availability

Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Allan C. Spradling (spradling@carnegiescience.edu).

Materials availability

This study did not generate new unique reagents.

Experimental model and subject details

Mouse strains used-C57BL/6, CAG-cre/ERT2 mice, R26R-EYFP (as described in [Lei and Spradling, 2016](#)) and Dazl1L^{+/-} mice (Strain #035880 from JACKSON laboratories). Genotyping was performed according to protocols provided by the JAX Genotyping was performed according to protocols from the JAX Mice database. Animals were provided with a proper light-dark cycle, temperature, humidity, food and water. Sexually mature females and males (6-8 weeks old) were kept for mating. Mating was confirmed through the observation of a vaginal plug. As per standard protocol, we designate the midday of the corresponding day as E0.5, which marks the beginning of embryonic development. Fetal gonads were dissected during E10.5-E18.5 as needed. The day pups are born is designated as P0 and were dissected for some experiments. The Institutional Animal care and use committee of Carnegie Institution of Washington approved procedures for animal handling and experimentation.

Method details

Single germ cell lineage labeling

The protocol used for single germ cell lineage labeling was performed as described previously ([Lei and Spradling, 2016](#)). Briefly, to obtain fetuses for lineage marking, adult female R26R-EYFP mice were mated with male CAG-cre/ERT2 mice. Tamoxifen was dissolved in corn oil (Sigma) and a single dose was injected intraperitoneally at 0.2 mg per 40 g body weight into pregnant female R26R-EYFP mice at E10.5. Fetal gonads/Pups were dissected as needed in between E11.5-E18.5/P0. To analyze the lineage-labeled germ cells, fixed and frozen whole gonads were subjected to whole

mount staining for chicken-GFP, the number of lineage-labeled germ cells of each ovary was determined by examining optically sectioned Z-stack images generated by confocal imaging of the entire ovary.

Immunostaining

Dissected fetal gonads/ovaries were washed in Phosphate buffer saline (PBS) and fixed immediately in cold 4% paraformaldehyde overnight at 4°C. Following the fixation, the gonads were washed three times with PBST₂, which is a mixture of phosphate-buffered saline (PBS) with 0.1 % Tween 20 (Sigma) and 0.5 % Triton X (Sigma). Each wash cycle lasted for 30 minutes. Next, gonads were incubated with primary antibodies mixed with Blocking solution (PBST₂ + 10 % Normal donkey serum) overnight at room temperature. Following day, gonads were incubated with fluorescein-conjugated secondary antibodies (Donkey Anti-Rabbit/mouse/rat Alexa-fluor 488/568/647, Invitrogen) overnight at room temperature. Next day, gonads were washed with PBST and stained with DAPI (Sigma) to visualize nuclei. Gonads were mounted on slides with a mounting medium (Vector Labs) and analyzed using confocal microscopy (STELLARIS 8 DIVE Multiphoton Microscope, Leica). Germ cell numbers are quantified in each ovary by manually counting EMA/DDX4 stained germ cells in multiple 0.45 µm optical sections from different ovaries.

3D reconstruction

Three-dimensional model images and movies were generated by Imaris software (Bitplane). For z-stack images, different visualization options in the Surpass mode of Imaris helped to gain visualization control of the objects. To perform surface rendering and volume quantification using Imaris software, we imported our 3D image dataset for EMA/WGA staining. We started with a “Volume” rendering for channels corresponding to EMA/WGA staining and then we adjusted the contrast, brightness and transparency in the “Display Adjustment” window. Next, we used the Surfaces module to create surface renderings by selecting the appropriate fluorescence channel, applied manual thresholding and adjusted smoothing parameters to define the structure accurately. Once the surface is rendered, we navigated to the statistics tab to access volume measurements. Further visualization of the results was performed in 3D or orthogonal views to confirm accuracy, and adjusted rendering properties such as color or transparency as needed. To create spots for easy counting of labelled cells, we used the “Spots” icon-option in the “Creation” menu and started the spot creation wizard. We set the approximate spot size as per the requirement. Next, we selected the “Manual Creation” option to place spots individually. Each click placed a single spot at the corresponding position.

Sexing and Genotyping

Tail samples from mouse were collected and subjected to lysis and DNA extraction using Proteinase K based lysis buffer treatment at 55°C overnight followed by 95°C incubation for 10 minutes. The samples were then centrifuged at 14000 rpm for 5 min to pellet the debris. Extracted genomic DNA from supernatant was used directly and amplified with the Uba1, Sly and Zfy primer pairs for sex determination (McFarlane et al., 2013 [DOI:10.1016/j.ydbio.2013.05.001](#), Table S2). PCR reactions were performed using KAPA Fast Hotstart ReadyMix with dye (Kappa Biosystems, #KK5608) using the following PCR parameters: initial denaturation at 95 °C for 5 min, 35 cycles with 94 °C for 30 s, 60° C and 72° C for 30 s, followed by final elongation at 72° C for 5 min. PCR products were analyzed with a DNA ladder (100 bp) on 2% agarose gels and visualized with ethidium bromide under UV-illumination. The male and female amplicon can be distinctly visualized due to different amplicon sizes of PCR products (See Primer details).

Electron microscopy

Whole ovaries were dissected in PBS and fixed in 4% Paraformaldehyde overnight. After three 3-min washes in cacodylate buffer, ovaries were postfixed in 1% OsO₄, 0.5% K₃Fe {CN}₆, in cacodylate buffer for 1 hour and were rinsed twice for 5 min in cacodylate buffer and once for 5 min with 0.05 M maleate (pH 6.0). The ovaries were stained in 0.5% uranyl acetate overnight at 4 degrees rinsed in water and dehydrated through an ethanol series. Following two 10-min washes

with propylene oxide, the ovaries were infiltrated with resin. The resin-embedded specimen was polymerized by incubation at 45⁰ C and 70⁰ C for 12 hours each. Silver-gold sections were cut, stained with lead citrate, and observed in the electron microscope.

Single cell RNA sequencing of Wildtype E10.5, E11.5 and E15.5 gonad

Embryonic ovaries were dissected such that individual fetal gonad at E10.5 (18 gonads from 3 females), E11.5 (12 female gonad from 3 females) and E15.5 (10 gonad from 2 females) were collected. For E10.5 and E11.5: corresponding tail were labelled for identification and placed in 1× PBS on ice. We performed quick Quinacrine and DAPI staining (~15 min) of corresponding E10.5 and E11.5 fetal tail samples to eliminate the male gonad samples -with positive Quinacrine staining of tail samples. We then pooled the dissected female gonad sample and dissociated it into single cells using 0.25% Trypsin at 37⁰ C for 5-7 min with two pipet trituration in between. Fetal bovine serum (10 %) was then used to neutralize trypsin. Dissociated cells were passed through 100-µm strainer. The cell suspension was centrifuged at 300 g for 5 minutes and cell pellet was resuspended in freshly prepared and filter-sterilized 0.04% BSA. Viability of cells was assessed via trypan staining and >90 percent viable samples were selected and loaded (~10000 live cells for E11.5 and E15.5, ~20000 cells for E10.5) onto the 10X Genomics Chromium Single Cell system using the for E11.5 and E15.5-v3 and for E10.5 v4 chemistry as per the manufacturer's instruction. Single-cell RNA capture and library preparations were performed, and standard data was data processing was performed using Cell ranger pipeline (6.0.1 - for E11.5 and E15.5 and 8.0.1-for E10.5) and later data was visualized and analyzed by Seurat v5.1.0.

Single cell RNA sequencing of Dazl1L mutant gonad

Dazl[±] female and males were kept for mating and plugged females were marked as E0.5. The gonad from E11.5, E12.5 fetuses were dissected, fetuses were collected in and kept in cold PBS in 12-well plate such that each well with individual fetus is marked for identification. Fetal tails were collected, labelled and subjected to kapa express extract kit (Catalog #KR0383- v4.16) mediated fast DNA extraction according to manufacturer's protocol. Briefly, one step lysis and DNA extraction system was set up in 100 µl volume by adding 10 µl express extract buffer, 2 µl of express extract enzyme and 88 µl of PCR grade water in each tail sample followed by 75⁰ C incubation in heating block for 15 minutes for lysis. The samples were then incubated at 95⁰C in heating block for 5 minutes for DNA extraction. The sample was then centrifuged briefly to pellet the debris, and supernatant was used directly for PCR as mentioned in methods section of RNA isolation, cDNA synthesis and PCR. The standard genotyping JAX protocol for strain 035880 is refer to identify homozygous / heterozygous and wild type fetuses. Simultaneously male/female sexing was also performed (primer details in Table S2). After correct identification, homozygous fetuses-from E11.5 (6 gonad from 3 females) and E12.5 (6 gonad from two females)- were trypsinized to prepare live single cells which were then subjected to 10X Genomics Chromium (v3 chemistry) and data was processed using cell ranger pipeline 6.0.1 and analyzed using seurat v5.1.0 in same manner as mentioned previously for E11.5 gonad.

Cell Identification and Clustering Analysis

Single-cell RNA sequencing (scRNA-seq) data were analyzed using the Seurat package (v5.1.0, <https://satijalab.org>). Count data generated by the Cell Ranger pipeline (6.0.1/8.0.1) were imported into R using the Read10X function and converted into a Seurat object with the CreateSeuratObject function. R packages were used to filter out the low-quality cells, and the following criteria were used to filter cells: **for E10.5 WT gonad:** nFeature_RNA > 500 & nFeature_RNA<8500 & nCount_RNA > 500 & percent.mt<5 **for E11.5 WT gonad:** nFeature_RNA > 100 & nFeature_RNA<11000 & nCount_RNA > 300 & percent.mt<10 **for E15.5 WT ovary:** nFeature_RNA > 500 & nFeature_RNA<9000 & nCount_RNA > 500 & percent.mt<10. The filtered count matrices were Log-normalized using the "NormalizeData" function and scaled with the "ScaleData" function to prepare for dimensionality reduction. "Principal Component Analysis (PCA)" was then applied to reduce dimensionality, followed by using top 15 dimensions and default

resolution to cluster cells based on gene expression profiles using the “FindNeighbors” and “FindClusters” functions. Cell populations were visualized using the umap method, facilitating the identification of distinct cell types.

Bioinformatic segregation of female germ cells

To segregate E10.5 and E11.5 XX-specific germ cells with complete surety we performed bioinformatic segregation of female germ cells to address the limitations of primitive Quinacrine staining. The expression of the X-linked gene *Xist*, *Ddx3x*, *Utx* etc., were used as a marker. Cells with significant *Xist* or other X-linked gene expression were labeled as “XXonly” using the “Idents” and “WhichCells” R functions. A new Seurat object containing only XX-specific cells was created using the subset function. Germ cells within this subset were identified by examining the expression of germ cell-specific marker genes, and a further subset of E10.5 and E11.5 female germ cells was created using the subset function to focus on germ cells specifically.

Wildtype ger cells-merged dataset creation and Validation

ScRNA-seq datasets from E11.5 to P5 (retrieved from the GEO database, GSE136441 and merged object submitted to Github) were visualized using Seurat. E11.5 cells were discarded and new merged dataset from E10.5-P5 was created, subsets of germ cell clusters from E10.5, E11.5, and E15.5 were integrated using Seurat’s “Merge” function. The resulting merged dataset was then processed by normalizing and scaling the data, followed by the identification of variable genes. The batch correction was performed using and pre-harmony and post-harmony data was assessed by visualizing LISI (Local Inverse Simpson’s Index) score confirming successful batch mixing. Cell clusters were determined using Seurat’s single nearest neighbor (SNN) algorithm with PCA action using top 15 dimensions and 2.8 resolution. To visualize these clusters, dimensionality reduction was applied (umap) with following criteria: `RunUMAP(new_merge, dims=1:15, n.neighbors=40, min.dist=0.6, spread=1)` enabling the identification of distinct cell populations.

Cell cycle stages of germ cells across the different developmental stages (E10.5-P5) were validated by visualizing meiosis stage specific expression pattern via feature plot. This visualization technique allows for the capture of subtle trends and helps in comparing expression peaks, providing insights into the transition from mitosis to meiosis across the different developmental stages. Additionally, a violin plot was employed to visualize the variability in average raw UMI levels across germ cells. This approach allowed for an effective comparison of gene expression profiles, highlighting the differences in gene expression patterns between different stages.

Dazl mutant merged dataset

For E11.5 and E12.5 *Dazl*^{-/-} dataset, cells were filtered based on following criteria:

- a) **for E11.5 *Dazl*^{-/-} ovary** : `nFeature_RNA > 500 & nFeature_RNA < 9000 & nCount_RNA > 500 & percent.mt < 5`
- b) **for E12.5 *Dazl*^{-/-} ovary** `nFeature_RNA > 500 & nFeature_RNA < 10000 & nCount_RNA > 500 & percent.mt < 5`

Top 15 dimensions were used at default resolution to cluster the cells which were then visualized by umap. To create merged dataset from subset containing only germ cells were created and merged using “merge” function as described previously. Three dataset were created -for E11.5 wildtype and *Dazl*^{-/-} dataset, for E12.5 and *Dazl*^{-/-} dataset and one dataset was created where both E11.5 and 12.5 WT and *Dazl*^{-/-} were merged. Batch correction was applied using harmony. Merged dataset was normalized, scaled and top 15 dimensions at resolution of 0.4 was used for visualization using umap.

In vitro gonad culture: microtubule inhibitor/BFA treatment and organelle tracker assay

To perform *in vitro* culture of mouse fetal ovaries on membrane inserts with or without inhibitors, we prepared sterile culture medium, constituting DMEM/F-12 supplemented with 10% fetal bovine serum and penicillin-streptomycin. Inhibitor stock solutions were prepared in DMSO and diluted at desired concentration in culture media (Ciliobrevin D from Millipore, 25 μ M for 6

hours, BFA from Invitrogen, 5 µg/ml for 6 hours). Control medium containing the vehicle DMSO was prepared simultaneously. Fetal ovaries were dissected from E11.5 embryos under a stereomicroscope in sterile PBS, we have kept the surrounding mesonephric tissue intact and attached to the gonad to ensure proper development. We have also collected corresponding fetus tail of each fetal gonad to perform genetic sex determination. We have placed membrane inserts into a 24-well plate and added ~500 µL of culture medium (with or without inhibitors) to the lower chamber and ensure no contact between medium and the insert surface. Next, we gently placed isolated ovaries onto the membrane surface using sterile forceps. The plate is then incubated in a CO₂ incubator at 37°C for ~6 hours. After the designated culture period, confirmed ovaries (post genetic-sex analysis), were carefully removed from the membrane inserts and washed three times with basal media to remove any residual inhibitors. Next, we collected ovaries and fixed in 4% PFA for downstream immunolocalization studies. To inhibit the microtubule growth using cold treatment, we transfer dissected gonad to a chilled culture medium, maintained at 0-4°C. The tissue is fully immersed in the pre-cooled medium and incubated on ice for ~60 minutes. This low-temperature exposure destabilizes microtubules by disrupting tubulin dynamics, effectively depolymerizing existing microtubules and inhibiting further polymerization. After the incubation, the tissue is washed with fresh ice-cold medium and fixed in 4% PFA. To stain specifically for organelles: Fetal ovaries were dissected and cultured *in vitro* in DMEM/F12 medium supplemented with organelle-specific fluorescent dyes. LysoTracker™ Deep (Thermo Fisher, L7528, 1 mM stock) or MitoTracker™ (Thermo Fisher, M7514) or MitoTracker™ Deep (Thermo Fisher, M22426), and ER-Tracker™ (Thermo Fisher, E34251) or ER-Tracker™ (Thermo Fisher, E34250) were each added at a final concentration according to the manufacturer's recommendations. The ovaries were incubated in this dye-containing medium at 37 °C with 5% CO₂ for 12 hours. Following incubation, tissues were collected and fixed in 4% paraformaldehyde (PFA) for subsequent immunostaining.

Magnetic activated cell sorting (MACS)

Embryonic ovaries were pooled and treated with Trypsin then were neutralized followed by centrifugation as mentioned before to in scRNA Sequencing method section to create dissociated single cells. The resulting cell pellet was re-suspended in ~80 microliters MACS buffer i.e. PBS with 0.5% BSA and 2 mM ethylenediaminetetraacetic acid (EDTA). ~20 microliters Anti-SSEA1 (CD15) microbeads (MiltenyiBiotec Inc., #130-094-530) were added to the cell suspension followed by incubated for 20 min on ice. After adding 1 ml MACS Buffer to the suspension, the SSEA1 bound cells were pelleted by centrifugation at 300 g for 10 min at 4°C. SSEA1+ve and SSEA1-ve cells were then separated by applying cell suspension to MS columns according to the manufacturer's instruction (MiltenyiBiotec Inc.). The SSEA1+ve cells were retained by the column and were filtered thrice to obtain pure germ cell population. The cells were counted, and viability was assessed via trypan staining and using automated cell counter (Fischer Scientific). For validation of MACS protocol the cells were fixed in 4 % PFA to perform Immunostaining of DDX4 and were also stored in trizol to perform RNA synthesis and PCR for germ cell specific gene expression.

IRE1-Xbp1 Assay

The IRE1-XBP1 ratiometric assay (from Montana Molecular #U0921G) is a genetically encoded biosensor system utilizes a BacMam Vector carrying dual-fluorescence biosensor. For each assay reaction 50 µl transduction mix consisting of XBP1-IRE1 sensor (15 µl) plus sodium butyrate (0.6 µl per well) was prepared in basal media. To standardize the working conditions, manufacturer provided thapsigargin (1 µM per well) was used as positive control and untransduced cells were used as negative control. For each experiment equal amounts of SSEA1- ve and SSEA1 +ve cells were added to a 96-well plate in duplicates. ~50 microliters of transduction mix was added to each reaction well. The plate was incubated for 45 minutes at room temperature and then transfer to 5% CO₂ and 37°C for 24 hrs. The enzymatic activity was then measured at 37°C by monitoring XBP1-IRE1 sensor's fluorescent intensity (top read) at Ex/Em=488/525 nm and basal constitutive fluorescent intensity at 565/620 using microplate reader (BioTek Synergy H1). IRE1-Xbp1 assay comparing MACS sorted SSEA1⁺ germ vs SSEA1⁻ somatic cells at E11.5 was performed across 6

experiments (~32 mice, ≥ 5 mice and ≥ 20 ovaries per experiment). IRE1-Xbp1 assay comparing SSEA1⁺ vs SSEA1⁻ E12.5 cells in WT or Dazl mutant mouse across 3 experiments (~40 mice, ≥ 5 mice and ≥ 25 ovaries per experiments)

20S Proteasome activity assay

To perform proteasome activity assay (Amplite #13456) we followed manufacturer's instructions. We first standardized the working conditions using trypsin enzyme as technical positive control. Trypsin is loaded on different well in different concentrations in duplicates. Blank medium along with assay buffer were added in separate wells to act as Negative control. For each experiment equal amounts of SSEA1-ve and SSEA1 +ve cells were added to a 96-well plate in duplicates. ~50 microliters of working solution consisting of fluorogenic substrate in assay buffer (prepared according to manufacturer's instructions) were added and plate was incubated for 2 hours at 37°C. The enzymatic activity was then measured at 37°C by monitoring fluorescent intensity (top read) at Ex/Em=490/525 nm (Cut off =515 nm) using microplate reader (BioTek Synergy H1). Proteasome activity in MACS-sorted SSEA1⁺ vs SSEA1⁻ cells were assessed using 3 biological assays with ~35–60 E11.5 ovary per assay. Proteasome activity ratio of MACS-sorted SSEA1⁺ and SSEA1⁻ cells compared at E12.5 with ~25–28 E12.5 ovaries were used per assay.

RNA isolation and cDNA synthesis

Total RNA was extracted using TRIzol reagent (Invitrogen) following the manufacturer's instructions. Briefly, tissue samples were homogenized in TRIzol, or cells were incubated in TRIzol for 5 min at RT, followed by phase separation with chloroform (200 μ l/ml TRIzol, shake for 10-15 seconds and keep at RT for 5 min) and centrifugation at 12,000 $\times$ g for 15 minutes at 4°C. The aqueous phase was collected, and RNA was precipitated with equal amount of isopropanol (10 min RT then same centrifugation as last step), washed with 75% ethanol, centrifuged, removed the supernatant ethanol and air-dried. Resuspended pellet in RNase-free water. RNA quality and concentration were determined spectrophotometrically. For cDNA synthesis we used Superscript IV cells Direct cDNA synthesis kit (Invitrogen, #11750150) with and modified manufacturer's protocol. We skipped the lysis step and directly added DNase I to our sample (0.5 μ l each) followed by 10 minutes incubation on ice. Then we added Stop solution (3 μ l each) and kept the mixture at RT for 2 minutes. Then we set up RT reaction as per manufacturer's instruction by adding 8 μ l of RT mix to each sample and setting up the following reaction in PCR thermocycler: 25⁰ C for 10 min, 55⁰ C for 10 min, 85⁰ C for 5 min and hold at 4⁰ C. cDNA was used directly for PCR as mentioned in Sexing and genotyping using DDX4 and GAPDH Primers (See Primer details).

Quantification and statistical analysis

Quantification of germ cell numbers

PFA fixed ovaries were subjected to whole mount immunostaining as described previously. Z-stack images for whole ovaries were analyzed using FIJI. Germ cell numbers are quantified in each ovary by manually counting EMA/DDX4 stained germ cells in multiple 0.45 μ m optical sections from different ovaries using FIJI.

3D volume Quantification

Whole mount immunostaining was performed and the z-stack images were converted to 3D image to analyze using Imaris as described previously. After using the Surfaces module to create surface rendering and applying manual thresholding and smoothing to define the structure accurately, we navigated to the statistics tab to access volume measurements. Further visualization of the results was performed in 3D or orthogonal views to confirm accuracy, and adjusted rendering properties such as color or transparency as needed. On an average the average values from three measurements performed were considered and plotted as graph using Graphpad prism.

Area of staining quantified using FIJI

Images were analyzed using FIJI to calculate area of staining by selecting region of interest and using measure command from Analyze tab. The results were plotted as number of pixels to depict the area corresponding to stained regions.

Statistics

Data are presented as mean $\pm$ SEM (standard error of the mean). Statistical analyses were performed using Student's paired t-test, one-way ANOVA with Tukey's post-hoc test, or the Friedman test with Dunn's post-hoc test for non-parametric data. Significance is indicated as follows: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.005$ (***), and $p < 0.001$ (****). Each experiment was performed on "N" individual specimens, as indicated in the figure legend, from at least three different animals.

Data availability

All datasets generated by this project are listed below. All other relevant materials are presented in the Figures, or the Supplement. Any additional materials not included are available on request.

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Additional files

[Supplemental Figures and video link](#)

Additional information

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Peer reviews

Reviewer #2 (Public review):

This study identifies Visham, an asymmetric structure in developing mouse cysts resembling the *Drosophila* fusome, an organelle crucial for oocyte determination. Using immunofluorescence, electron microscopy, 3D reconstruction, and lineage labeling, the authors show that primordial germ cells (PGCs) and cysts, but not somatic cells, contain an EMA-rich, branching structure that they named Visham, which remains unbranched in male cysts. Visham accumulates in regions enriched in intercellular bridges, forming clusters reminiscent of fusome "rosettes." It is enriched in Golgi and endosomal vesicles and partially overlaps with the ER. During cell division, Visham localizes near centrosomes in interphase and early metaphase, disperses during metaphase, and reassembles at spindle poles during telophase before becoming asymmetric. Microtubule depolymerization disrupts its formation.

Cyst fragmentation is shown to be non-random, correlating with microtubule gaps. The authors propose that 8-cell (or larger) cysts fragment into 6-cell and 2-cell cysts. Analysis of *Pard3* (the mouse ortholog of *Par3/Baz*) reveals its colocalization with Visham during cyst asymmetry, suggesting that mammalian oocyte polarization depends on a conserved system involving *Par* genes, cyst formation, and a fusome-like structure.

Transcriptomic profiling identifies genes linked to pluripotency and the unfolded protein response (UPR) during cyst formation and meiosis, supported by protein-level reporters monitoring *Xbp1* splicing and 20S proteasome activity. Visham persists in meiotic germ cells

at stage E17.5 and is later transferred to the oocyte at E18.5 along with mitochondria and Golgi vesicles, implicating it in organelle rejuvenation. In *Dazl* mutants, cysts form, but Visham dynamics, polarity, rejuvenation, and oocyte production are disrupted, highlighting its potential role in germ cell development.

Overall, this is an interesting and comprehensive study of a conserved structure in the germline cells of both invertebrate and vertebrate species. Investigating these early stages of germ cell development in mice is particularly challenging. Although primarily descriptive, the study represents a remarkable technical achievement. The images are generally convincing, with only a few exceptions.

Major comments:

(1) Some titles contain strong terms that do not fully match the conclusions of the corresponding sections.

(1a) Article title "Mouse germline cysts contain a fusome-like structure that mediates oocyte development":

The term "mediates" could be misleading, as the functional data on Visham (based on comparing its absence to wild-type) actually reflects either a microtubule defect or a *Dazl* mutant context. There is no specific loss-of-function of *visham* only.

(1b) Result title, "Visham overlaps centrosomes and moves on microtubules":

The term "moves" implies dynamic behavior, which would require live imaging data that are not described in the article.

(1c) Result title, "Visham associates with Golgi genes involved in UPR beginning at the onset of cyst formation":

The presented data show that the presence of Visham in the cyst coincides temporally with the expression and activity of the UPR response; the term "associates" is unclear in this context.

(1d) Result title, "Visham participates in organelle rejuvenation during meiosis":

The term "participates" suggests that Visham is required for this process, whereas the conclusion is actually drawn from the *Dazl* mutant context, not a specific loss-of-function of *visham* only.

(2) The authors aim to demonstrate that Visham is a fusome-like structure. I would suggest simply referring to it as a "fusome-like structure" rather than introducing a new term, which may confuse readers and does not necessarily help the authors' goal of showing the conservation of this structure in *Drosophila* and *Xenopus* germ cells. Interestingly, in a preprint from the same laboratory describing a similar structure in *Xenopus* germ cells, the authors refer to it as a "fusome-like structure (FLS)" (Davidian and Spradling, *BioRxiv*, 2025).

Comments on revisions:

The revised manuscript has been clearly improved, and the authors have addressed all of our comments. I would like to point out two minor issues:

(1) As suggested by the reviewers, the authors now use the term fusome instead of *visham*. However, they also acknowledge that this structure lacks many components of the *Drosophila* fusome. It may therefore be more appropriate to refer to it as a "mouse fusome" or as a "fusome-like structure (FLS)," as used in *Xenopus*.

(2) I agree with Reviewer 3 that co-localization between EMA and acTubulin on still images does not convincingly demonstrate that fusome vesicles move along microtubules (Figure S2E).

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Reviewer #3 (Public review):

The manuscript provides evidence that mice have a fusome, a conserved structure most well studied in *Drosophila* that is important for oocyte specification. Overall, a myriad of evidence is presented demonstrating the existence of a mouse fusome. This work is important as it addresses a long-standing question in the field of whether mice have fusomes and sheds light on how oocytes are specified in mammals.

Comments on revisions:

Overall, the authors did a good job of responding to reviewer comments that have improved the manuscript by including higher quality microscope images, revising text for clarity and using the term mouse fusome instead of using a new term. However, two of the headings in the results section that didn't correspond to the data presented in that section still have not been revised even though the authors stated that they were revised in their response to reviewer comments. The heading of the first section of the results is: "PGCs contain a Golgi-rich structure known as the EMA granule" even though no evidence in that section shows it is Golgi rich. The heading of the fifth section of the results is: "The mouse fusome associates with polarity and microtubule genes including *pard3*" however, only evidence for *pard3* is presented.

<https://doi.org/10.7554/eLife.109358.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review)

Summary

We thank the reviewer for the constructive and thoughtful evaluation of our work. We appreciate the recognition of the novelty and potential implications of our findings regarding UPR activation and proteasome activity in germ cells.

(1) The microscopy images look saturated, for example, Figure 1a, b, etc. Is this a normal way to present fluorescent microscopy?

The apparent saturation was not present in the original images, but likely arose from image compression during PDF generation. While the EMA granule was still apparent, in the revised submission, we will provide high-resolution TIFF files to ensure accurate representation of fluorescence intensity and will carefully optimize image display settings to avoid any saturation artifacts.

(2) The authors should ensure that all claims regarding enrichment/lower vs. lower values have indicated statistical tests.

We fully agree. In the revised version, we will correct any quantitative comparisons where statistical tests were not already indicated, with a clear statement of the statistical tests used,

including p-values in figure legends and text.

(a) In Figure 2f, the authors should indicate which comparison is made for this test. Is it comparing 2 vs. 6 cyst numbers?

We acknowledge that the description was not sufficiently detailed. Indeed, the test was not between 2 vs 6 cyst numbers, but between all possible ways 8-cell cysts or the larger cysts studied could fragment randomly into two pieces, and produce by chance 6-cell cysts in 13 of 15 observed examples. We will expand the legend and main text to clarify that a binomial test was used to determine that the proportion of cysts producing 6-cell fragments differed very significantly from chance.

Revised text:

“A binomial test was used to assess whether the observed frequency of 6-cell cyst products differed from random cyst breakage. Production of 6-cell cysts was strongly preferred (13/15 cysts; **** $p < 0.0001$).”

(b) Figures 4d and 4e do not have a statistical test indicated.

We will include the specific statistical test used and report the corresponding p-values directly in the figure legends.

(3) Because the system is developmentally dynamic, the major conclusions of the work are somewhat unclear. Could the authors be more explicit about these and enumerate them more clearly in the abstract?

We will revise the abstract to better clarify the findings of this study. We will also replace the term Visham with mouse fusome to reflect its functional and structural analogy to the Drosophila and Xenopus fusomes, making the narrative more coherent and conclusive.

(4) The references for specific prior literature are mostly missing (lines 184-195, for example).

We appreciate this observation of a problem that occurred inadvertently when shortening an earlier version. We will add 3–4 relevant references to appropriately support this section.

(5) The authors should define all acronyms when they are first used in the text (UPR, EGAD, etc).

We will ensure that all acronyms are spelled out at first mention (e.g., Unfolded Protein Response (UPR), Endosome and Golgi-Associated Degradation (EGAD)).

(6) The jumping between topics (EMA, into microtubule fragmentation, polarization proteins, UPR/ERAD/EGAD, GCNA, ER, balbiani body, etc) makes the narrative of the paper very difficult to follow.

We are not jumping between topics, but following a narrative relevant to the central question of whether female mouse germ cells develop using a fusome. EMA, microtubule fragmentation, polarization proteins, ER, and balbiani body are all topics with a known connection to fusomes. This is explained in the general introduction and in relevant subsections. We appreciate this feedback that further explanations of these connections would be helpful. In the revised manuscript, use of the unified term mouse fusome will also help connect the narrative across sections. UPR/ERAD/EGAD are processes that have been studied in repair and maintenance of somatic cells and in yeast meiosis. We show that the major regulator XbpI is found in the fusome, and that the fusome and these rejuvenation pathway genes are expressed and maintained throughout oogenesis, rather than only during limited late stages as suggested in previous literature.

(7) The heading title "Visham participates in organelle rejuvenation during meiosis" in line 241 is speculative and/or not supported. Drawing upon the extensive, highly rigorous *Drosophila* literature, it is safe to extrapolate, but the claim about regeneration is not adequately supported.

We believe this statement is accurate given the broad scope of the term "participates." It is supported by localization of the UPR regulator XbpI to the fusome. XbpI is the ortholog of HacI a key gene mediating UPR-mediated rejuvenation during yeast meiosis. We also showed that rejuvenation pathway genes are expressed throughout most of meiosis (not previously known) and expanded cytological evidence of stage-specific organelle rejuvenation later in meiosis, such as mitochondrial-ER docking, in regions enriched in fusome antigens. However, we recognize the current limitations of this evidence in the mouse, and want to appropriately convey this, without going to what we believe would be an unjustified extreme of saying there is no evidence.

Reviewer #2 (Public review):

We thank the reviewer for the comprehensive summary and for highlighting both the technical achievement and biological relevance of our study. We greatly appreciate the thoughtful suggestions that have helped us refine our presentation and terminology.

(1) Some titles contain strong terms that do not fully match the conclusions of the corresponding sections.

(1a) Article title "Mouse germline cysts contain a fusome-like structure that mediates oocyte development"

We will change the statement to: "Mouse germline cysts contain a fusome that supports germline cyst polarity and rejuvenation."

(1b) Result title "Visham overlaps centrosomes and moves on microtubules"

We acknowledge that "moves" implies dynamics. We will include additional supplementary images showing small vesicular components of the mouse fusome on spindle-derived microtubule tracks.

(1c) Result title "Visham associates with Golgi genes involved in UPR beginning at the onset of cyst formation"

We will revise this title to: "The mouse fusome associates with the UPR regulatory protein Xbp1 beginning at the onset of cyst formation" to reflect the specific UPR protein that was immunolocalized.

(1d) Result title "Visham participates in organelle rejuvenation during meiosis"

We will revise this to: "The mouse fusome persists during organelle rejuvenation in meiosis."

(2) The authors aim to demonstrate that Visham is a fusome-like structure. I would suggest simply referring to it as a "fusome-like structure" rather than introducing a new term, which may confuse readers and does not necessarily help the authors' goal of showing the conservation of this structure in *Drosophila* and *Xenopus* germ cells. Interestingly, in a preprint from the same laboratory describing a similar structure in *Xenopus* germ cells, the authors refer to it as a "fusome-like structure (FLS)" (Davidian and Spradling, BioRxiv, 2025).

We appreciate the reviewer's insightful comment. To maintain conceptual clarity and align with existing literature, we will refer to the structure as the mouse fusome throughout the manuscript, avoiding introduction of a new term.

Reviewer #3 (Public review):

We thank the reviewer for emphasizing the importance of our study and for providing constructive feedback that will help us clarify and strengthen our conclusions.

(1) Line 86 - the heading for this section is "PGCs contain a Golgi-rich structure known as the EMA granule"

We agree that the enrichment of Golgi within the EMA PGCs was not shown until the next section. We will revise this heading to:

"PGCs contain an asymmetric EMA granule."

(2) Line 105-106, how do we know if what's seen by EM corresponds to the EMA1 granule?

We will clarify that this identification is based on co-localization with Golgi markers (GM130 and GS28) and response to Brefeldin A treatment, which will be included as supplementary data. These findings support that the mouse fusome is Golgi-derived and can therefore be visualized by EM. The Golgi regions in E13.5 cyst cells move close together and associate with ring canals as visualized by EM (Figure 1E), the same as the mouse fusomes identified by EMA.

(3) Line 106-107-states "Visham co-stained with the Golgi protein Gm130 and the recycling endosomal protein Rab11a1". This is not convincing as there is only one example of each image, and both appear to be distorted.

Space is at a premium in these figures, but we have no limitation on data documenting this absolutely clear co-localization. We will replace the existing images with high-resolution, noncompressed versions for the final figures to clearly illustrate the co-staining patterns for GM130 and Rab11a1.

(4) Line 132-133---while visham formation is disrupted when microtubules are disrupted, I am not convinced that visham moves on microtubules as stated in the heading of this section.

We will include additional supplementary data showing small mouse fusome vesicles aligned along microtubules.

(5) Line 156 - the heading for this section states that Visham associates with polarity and microtubule genes, including *pard3*, but only evidence for *pard3* is presented.

We agree and will revise the heading to: "Mouse fusome associates with the polarity protein *Pard3*." We are adding data showing association of small fusome vesicles on microtubules.

(6) Lines 196-210 - it's strange to say that UPR genes depend on DAZ, as they are upregulated in the mutants. I think there are important observations here, but it's unclear what is being concluded.

UPR genes are not upregulated in DAZ in the sense we have never documented them increasing. We show that UPR genes during this time behave like pluripotency genes and normally decline, but in DAZ mutants their decline is slowed. We will rephrase the paragraph to clarify that *Dazl* mutation partially decouples developmental processes that are normally linked, which alters UPR gene expression relative to cyst development.

(7) Line 257-259-wave 1 and 2 follicles need to be explained in the introduction, and how these fits with the observations here clarified.

Follicle waves are too small a focus of the current study to explain in the introduction, but we will request readers to refer to the cited relevant literature (Yin and Spradling, 2025) for further details.

We sincerely thank all reviewers for their insightful and constructive feedback. We believe that the planned revisions—particularly the refined terminology, improved image quality, clarified statistics, and restructured abstract—will substantially strengthen the manuscript and enhance clarity for readers.

Reviewer #1 (Recommendations for the authors):

(1) Figure 1E: need to use some immuno-gold staining to identify the Visham. Just circling an area of cytoplasm that contains ER between germ cell pairs is not enough.

We appreciate the reviewer's insistence that the association between the mouse fusome and Golgi be clearly demonstrated. However, the EMA granule is a large structure discovered and defined by light microscopy, and presents no inherent challenge to documenting its Golgi association by immunofluorescence experiments, which we presented and now further strengthened as described in the next paragraph. We believe that the suggested EM experiment would add little to the EM we already presented (Figure 1E, E') Moreover, due to facility limitations, we are currently unable to perform immunogold staining.

To strengthen previous immunolocalization experiments, we have now included additional immunostaining data showing the clear colocalization of the fusome region with the Golgi markers GM130 and GS28 (Figure S1H). We have also incorporated a new experiment using the Golgi-specific inhibitor Brefeldin A (BFA) see Figure S1I. Treatment of in vitro-cultured gonads with BFA, disrupted EMA granule formation, demonstrating that EMA granules not only associate with Golgi, but require Golgi function to to be maintained.

Additionally, in Figure 2, we showed that the fusome overlaps with the peri-centriolar region—a characteristic locus for Golgi due to its movement on microtubules. We showed that the dynamic behavior of the fusome during the cell cycle, parallels Golgi dispersal and reassembly, and all these facts provide further strong support for the Golgi-association of the EMA granule and fusome.

(2) Figure 1F: is this image compressed?

We have now substituted the image in Figure 1F with a better image and have avoided the compression of the image.

(3) In the figure legends, are the sample sizes individual animals or individual sections? Please ensure that all figure legends for each figure panel consistently contain the sample size.

We have now included the number of measurements (N) in every figure legend. Each experiment was performed using samples from at least three different animals, and in most cases from more than three. This information has also been added to the Methods section under Statistics. In addition, N values are now consistently provided for each graph throughout the figures.

(4) Figure 2b/c: seemly likely based on the snapshot of different stages of cytokinesis that the "newly formed" visham is accurate, but without live imaging, this claim of "newly formed" is putative/speculative. It is OK if it is labeled as "putative" in the figure panel.

The behavior of the *Drosophila* fusome during mitosis was deduced without live imaging (deCuevas et al. 1998). We clarified that the conversion of a single mouse germ cell with one round fusome to an interconnected pair of cells with two round fusomes of greater total

volume following mitosis is the basis for deducing that new fusome formation occurs each cell cycle. However, we agree with the reviewer that the phrase "newly formed" in the original label on Figure 2c suggested a specific mechanism of fusome increase that was not intended and this phrase has been removed entirely.

(5) Figure 2e/e is extremely difficult to follow. In order to improve the readability of these figure panels, can individual panels with a single stain be shown? The 'gap' between YFP+ sister cells is not immediately obvious in panel e or e" with the current layout. Since this is a key aspect of the author's claim about cleavage of the cyst, it would be best to make this claim more robust by showing more convincing images. In Figure 2E, the staining pattern of EMA needs to be clarified and described more fully in the text.

We mapped discontinuities in the microtubule connections, not the fusome or YFP. YFP is the lineage marker indicating that the cells of a single cyst are being studied. Consequently, no gap between YFP cytoplasmic expression is expected because only in the last example (figure E"), has fragmentation already occurred (and here there is a YFP gap). The acetylated tubulin gap proceeds fragmentation. The mitotic spindle remnants labeled by AcTub link the cells into two groups separated by a gap, which is clearly shown in the data images and in the third column where only the relevant AcTub from the cyst itself is shown. In response to the reviewers question about the fusome, which is not directly relevant to fragmentation, we have now provided images of the separate fusome channel and corresponding measurements for all three Figure 2E-E" cysts in the supplementary Figure S4H. We have improved the text regarding this important figure to try and make it easier to follow, and also added a new example of a 10-cell cyst also in S2H (lower panels). We also added, movies allowing full 3D study of one of the 8 cell cysts and the new 10-cell cyst. I also suggest that the reviewer examine how the deduced mechanism of fragmentation explains previously published but not fully understood data on cyst fragmentation going back to 1998 as described in the expanded Discussion on this topic.

(6) It would be best to support the proposed model in Figure 2G (4+4+4) with microscopy images of a 12-cell or 16-cell cyst? Would these 12-cell or 16-cell cysts be too large to technically recover in a section?

Unfortunately the reviewer 's suggestion that 12- or 16-cell cysts are too large to recover and present convincingly is correct. Because our analysis depends on capturing lineage-labeled cysts specifically at telophase with acetylated-tubulin connections, the likelihood of obtaining the correct stage is very low. In addition, the dense packing of germ cells in the mouse gonad further limits our ability to fully reconstruct all the cells in large cysts, with difficulty increasing as cyst size grows.

However, as noted, we added a well-resolved 10-cell cyst—the largest size we could confidently analyze—in a 3D video in Supplementary Figure S2H (lower panel), which shows a 6 + 4 breakage pattern.

(7) We did not find a reference in the text for Figure 2G.

We have now provided reference for 2G in the text and in the discussion section.

(8) Line 189: ERAD is used as an acronym, but is not defined until the discussion.

We have now provided full form of acronym at its first usage in the text.

(9) Fig 3i/i': the increase of UPR pathway components, increasing expression during zygote, is interesting to note, but is not commented enough in the text of the paper.

We have discussed this issue in the discussion section with specific reference to figure 3I. Please find the detailed discussion under the heading "Germ cell rejuvenation is highly active

during cyst formation.”

(10) Please quantify DNMT3A expression levels in WT control vs Dazl KO germ cells in Figure 4a.

We have now quantified DNMT3A expression levels in WT control vs Dazl KO germ cells and have added the data in the Figure 4A.

(11) Please introduce the rationale behind selecting Dazl KO for studying cyst formation (text in line 197). This comes out of nowhere.

True. We significantly expanded our discussion of Dazl and citations of previous work, including evidence that it can affect cyst structures like ring canals, in the Introduction.

(12) It would be best to stain WT control vs DazL KO oogonia in Figure 4a with 5mC antibodies to support their claim that DNA methylation might be affected in the mutants.

We respectfully disagree that this additional experiment is necessary within the scope of the current study. At the developmental stage examined (E12.5), germ cells in the Dazl mutant are clearly in an arrested and hypomethylated state, as supported by previous evidence (Haston et al. 2009). This initial experiment was designed to show that in our hands Dazl mutants show this known pluripotency delay. However, the effects of Dazl mutation on female germline cyst development as it relates to polarity or the fusome was not studied before, and that is what the paper addresses, building on previous work.

Because our study does not focus on germ-cell epigenetic modifications but rather on the consequences of Dazl loss on germ cell cyst development, adding 5mC immunostaining would not substantially advance the main conclusions. The existing data and previous published work already provide sufficient background.

(13) Figure 4c: a very interesting figure, it would be best to quantify developmental pseudotime (perhaps using monocle3 analysis) and compare more rigorously the developmental stage of WT control vs DazL KO.

Developmental pseudotime, such as through Monocle3 analysis, might sometimes be valuable but involves assumptions that when possible are better addressed by direct experimental examination. Our conclusions regarding cyst developmental stage are supported by straightforward evidence rather than to which computational trajectory inference would add little. Specifically, we have performed analysis of germ-cell methylation state, ring canal formation, pluripotency markers, UPR pathway activity assay (Xbp1 and Proteomic assay), Golgi-stress analysis and Pard3 which collectively document the developmental status of the WT and Dazl KO germ cells. These empirical data demonstrate the same developmental pattern reflected in Figure 4c, making the less reliable pseudotime-based computational method superfluous.

(14) Figure 4d has two panels labeled as "d".

We have now corrected the labelling of the figure

(15) Color coding in 4d, d', d" is confusing; please harmonize some visual presentation here.

We have now harmonized the visual representation of all the graphs in figure 4

(16) Fig 4e' is labeled as DazL +/- but is this really a typo?

Thank you for pointing it out. We have now corrected the typo

(17) Figure F: typo labeled as E3.5, which is E13.5?

Thank you for pointing it out. We have now corrected the typo

(18) *Figure F: was DazL KO mutant but no WT control.*

The WT control was not provided to avoid the redundancy. Please refer to earlier figure 3A-B, Fig S3C and D and videos S3A and S3b to refer to WT control at every stage.

(19) *Figure G: unusual choice in punctuation marks for cartoon schematic. No key to guide the reader for color-coded structures would be helpful to have something similar to 4h.*

We have now provided the key to guide the readers in the mentioned figure 4G.

(20) *The authors use WGA and EMA as interchangeable markers (Figure 5a) without fully explaining why they have switched markers.*

Because it is germ cell specific, we used EMA as a fusome marker during the time when it is found up through E13.5. After that point we used WGA which is still usable, but also labels somatic cells. This rationale is explicitly described at the end of the section “Fusome is highly enriched in Golgi and vesicles”, where we state:

“EMA staining disappears from germ cells at E14.5 (Figure 1I). However, very similar (but non-germ-cell-specific) staining continued with wheat germ agglutinin (WGA) at later stages (Figure 1G, G’; Figure S1G).”

To ensure this is fully clear to readers, we have now added an additional statement in the start of the text section discussing the figure 5:

“For the reasons explained previously (see text for Figure 1G), WGA was used as a fusome marker beyond stage E14.5.”

(21) *Figure 5b' is compressed.*

We have now decompressed the image

(22) *Line 267, Balbiani body is misspelled.*

We have now corrected the spelling.

(23) *The explanation of why the authors switch focus from DazL KO to DazL +/- is not adequately described. The authors should also explain the phenotype of the DazL +/- animals or reference a paper citing the hets are sterile or subfertile.*

We have now added the explanation of why Dazl KO is used in our introduction section where we have mentioned the phenotype of Dazl homozygous and heterozygous mouse.

(24) *Is Figure 5i actually DazL +/-? It is not labeled clearly in the text, the figure legend, or the figure panel.*

We have now labelled the figure correctly in figure and in the legend.

(25) *The paper ends abruptly at line 275 with no context or summary.*

The manuscript does not end at line 275; the apparent interruption is due to a page break occurring immediately before the beginning of the Discussion section. We hope that continuation is fully visible in the reviewer 1 (your) version of the PDF.

Reviewer #2 (Recommendations for the authors):

(1) Line 93: Fig. 1B: DDX4 marks germ cells; do all the red and yellow cells in the NE inset originate from the same PGC? There are only 2 cells marked in yellow among the group of red cells. Is it a z-projection issue? Or do they come from different PGCs?

This experiment used vasa staining to identify all germ cells, which are produced by multiple PGCs. Green labeling is a lineage marker derived from a single PGC (due to the low frequency of tamoxifen-activated labeling). Consequently, the two yellow cells observed in the NE inset of Fig. 1B represent YFP-labeled germ cells (YFP + DDX4 double-positive) that have arisen from a single, lineage-traced PGC. This approach, introduced in 2013, is described in the Methods, and represents the field's single largest technical advance that has made it possible to analyze mouse germ cell development at single cell resolution.

To ensure clarity, we have added a brief explanatory note to the figure legend indicating that yellow cells represent the lineage-traced progeny of a single PGC, while the red staining marks all germ cells.

(2) Line 96: Figure 1C vs 1C'. The difference between female and male Visham is not obvious, although quantification shows a clear difference. How was the quantification made? Manual or automatic thresholding? Would it be possible to show only the Visham channel?

We thank the reviewer for pointing out this problem. We have now more clearly described in the text that the female fusome increases in some cells with close attachments to other cells (future oocytes) and decreases in distant nurse cells. It branches due to rosette formation.. In males, the fusome remains much like the initial EMA granules present in early germ cells, with only fine and difficult to see connections. The quantification shown in Figures 1C and 1C' was performed manually, based on the presence of either (i) fused, branched EMA-positive fusome structures or (ii) dispersed, punctate EMA granules. This assessment was carried out across multiple E13.5 male and female gonad samples to ensure robustness. To facilitate independent evaluation, we have already provided supplementary videos S3B1 and S3B2, which display the EMA-stained E13.5 male and female gonads in three dimensions. These videos allow the structural differences to be examined more clearly than in static images.

In response to the reviewer's request, we now additionally include the single-channel fusome image in Supplementary Figure S1E'. This presentation highlights the fusome signal alone and further clarifies the morphological differences underlying the quantification.

(3) L118: Figure 2A, third row = 2-cell cyst? Please specify PCNT in the legend.

We appreciate the reviewer's observation. In Figure 2A (third row), the cells were not specifically labeled as a 2-cell cyst; rather, the intention was to illustrate the presence of two distinct centrosomes positioned on a fused fusome structure, a configuration we frequently observe.

We have now updated the figure legend to explicitly define PCNT.

(4) L169: Missing reference to S3B and video S3B1?

We have now included the reference to S3B1 and S3B2 in the text and in the legend

(5) L170: Please describe the graph in the Figure 3D legend.

We have now described the Graph in the legend

(6) L171: Would it be possible to have a close-up showing both Pard3 and Visham in a ringlike pattern related to RACGAP (RC) staining? The images are too small.

It is difficult to capture this relationship perfectly in a two dimensional picture. The images represent the maximum close-up possible that still includes enough relevant area for the necessary conclusions. We have now provided additional three close-up images exclusively for ring-canal and Pard3 association in the supplementary Figure S3C for further clarity. However, we also note that the quality of the image permits the reader of a pdf to zoom and to visualize the images in great detail.

| (7) L181: *Wrong reference, should be 3 then 3I.*

Thank you for pointing it out, we have now corrected the reference.

| (8) L199: *In Figure S4B, was DNMT3 staining quantified? Red intensity differs globally between images; use the somatic red level as a reference? Note: EMA seems higher in Dazl- vs. WT?*

We have now performed quantification of DNMT3 staining, which is presented in Figure 4A. While the red intensity (DNMT3 or EMA) can appear to differ between images, this variation can result from biological differences between tissues or minor technical variability despite using consistent microscope settings. To account for this, we normalized the staining intensity using the somatic cell signal as an internal reference, ensuring that the quantification reflects genuine differences between WT and Dazl-/- samples rather than global intensity variation.

| (9) L229: *Should be "proteasome."*

We have now corrected the spelling error.

| (10) L233: *Quantify fragmentation of Gs28? EMA doesn't seem affected. Could you quantify both Gs28 and EMA? Images are too small.*

We thank the reviewer for this suggestion. While the current images are small, they can be examined in detail using zoom to visualize the structures clearly. As noted, EMA staining is not affected, (we agree) as cells are in arrested state. This arrested state creates stress on Golgi. The fragmentation of Gs28-labeled Golgi membranes is a classical indicator of Golgi stress, even though the fragmented membranes may remain functionally active. Our results show that Dazl deletion specifically affects Golgi in germ cells, while Golgi in neighboring somatic cells appears healthy. To quantify this effect, we have now included manual quantification of Golgi fragmentation in Figure 4F, assessing tissues for the presence of fragmented versus intact Golgi structures. This confirms that Golgi fragmentation is a germ cell-specific phenotype in Dazl-/- samples, while pre-formed EMA-positive fusomes remain unaffected but probably in arrested state.

| (11) L237: *Figure 4F graph shows E3.5, not E13.5.*

We have now corrected the typo in the figure

| (12) L257: *Figure 5D: quantify as in 5A? overlap?*

Yes, it's an overlap and shown as two separate image with ring canal for better clarity. We have now quantified the image and have produced combined graph for fusome and pard3 in Figure 5A graph.

| (13) L261: *Figure 5E-E': black arrowhead not mentioned in legend.*

We have now mentioned the black arrowhead in the legend

| (14) L262: *Figure 5C: arrowhead not mentioned in legend. Figure 5F: oocyte appears separated from nurse cells compared to 5C.*

Yes, that may happen as cysts undergo fragmentation; what matters is all cells are lineage labelled and hence are members of a single cyst derived from one PGC.

| (15) L263: *Figure 5G has no legend reference; nurse cells are not outlined as in 5C.*

We have now outlined the nurse cells and have added the reference to the graph in the legend.

| (16) L279: *"The fusome and Visham and both..." should be replaced with "Both fusome and Visham..."*.

We have now replaced the term Visham with fusome as suggested by reviewers and editor. We updated the statement to correct the grammatical error.

| (17) L1127: *Video S3B1: It is unclear what to focus on.*

We have now added the Rectangle area and arrow to highlight what to focus on

| (18) L1128: *Video "S3B1" should be "S3B2."*

We have now corrected the legend

| (19) *Finally: curiosity question: have the authors tried to use known markers of the Drosophila fusome in mice, such as Spectrin or other markers described in Lighthouse, Buszczak and Spradling, Dev Bio, 2008? And conversely, do EMA and WGA label the fusome in Drosophila?*

Yes, we and others used the most specific markers of the *Drosophila* fusome such alpha-spectrin, adducin-like Hts, tropomodulin, etc. to search for fusomes in vertebrate species. It was unsuccessful in clarifying the situation, because Hts and alpha-spectrin in *Drosophila* and other insects generate a protein skeleton that stabilizes the fusome and is easily stained. But this structure is simply not conserved in vertebrates. The polarity behavior of the fusome, its core developmental property, is conserved, however. The mammalian fusome still acquires and maintains cyst polarity, and goes even farther and reflects both initial cyst formation and cyst cleavage, before marking oocyte vs nurse cell development in the smaller cysts. Expression of the inner microtubule-rich portion of the fusome, its Par proteins, and many ER-related and lysosomal fusome proteins are mostly conserved but their ability to mark the fusome alone varies with time and context (only some of the examples are shown in Figure 3I'). Nearly all of the proteins identified in Lighthouse et al. 2008 are expressed. These proteins may be involved in rejuvenation as studied here. We modified the first section of the Discussion to explicitly compare mouse, *Xenopus* and *Drosophila* fusomes, which was not possible before this work.

Reviewer #3 (Recommendations for the authors):

The authors should either revise the conclusions or add additional evidence to support their claims. In addition, minor corrections are listed below.

We have added additional evidence as noted in responses above, and revised some claims that were stated inaccurately. In addition, we have attempted to clarify the evidence we do present, so that its full significance is more easily grasped by readers.

| (1) *Lines 20-21 are unclear - the cyst doesn't get sent into meiosis, each oocyte does.*

Research is showing that it's more complicated than that. All cyst cells enter "pre-meiotic S phase", and most cell cycles are conventionally considered to start after the previous M phase-

i.e. in G1 or S, not in the next prophase, an ancient view limited just to meiosis. Absent this old tradition from meiosis cytology, pre-meiotic S would just be called meiotic S as some workers on meiosis do. In addition, in different species, nurse cells diverge from meiosis on different schedules, including many much later in the meiotic cycle. Two cyst cells in *Drosophila* fully enter meiosis by all criteria, the oocyte and one nurse cell that only exits in late zygotene. In *Xenopus* and mouse, scRNAseq shows that many cyst cells enter meiosis up to leptotene and zygotene, including nurse cells that specifically downregulate meiotic genes during this time, possibly to assist their nurse cell functions, while others remain in meiosis even longer (Davidian and Spradling, 2025; Niu and Spradling, 2022). Eventually, only the oocytes within each fragmented mouse cyst complete meiosis.

(2) Many places in the manuscript abbreviations are never defined or not defined the first time they are used (but the second or third time): Line 23-ER, Line 29-UPR, Line 33-PGC (not defined until line 45), Line 79-EGAD.

We have defined full acronyms now upon their first occurrence.

(3) Line 5 should be the pachytene substage of meiosis I.

We have now updated the statement to “In pachytene stage of meiosis I...”

(4) Line 59-61 - this statement needs a reference(s).

These statements are a continuation from the references cited in the previous statements. However, for further clarity we have again cited the relevant reference here (Niu and Spradling, 2022).

(5) Line 80 - should it be oocyte proteome quality control?

We have now updated the statement to “Oocyte proteome quality control begins early”.

(6) Line 87 - in this case, EMA does not stand for epithelial membrane antigen (AI will call it that, but it is not correct). I believe it originally was the abbrev for (Em)bryonic (a)ntigen, though some papers call it (e)mbryonic (m)ouse (a)ntigen. And the reference here is Hahnel and Eddy, 1986, but in the reference list is a different paper, 1987 (both refer to EMA-1).

We have now updated the acronym EMA-1 in corrected form and have corrected the citation.

(7) Line 176 - RNA seq.

We have now updated the statement to “We performed single cell RNA sequencing (scRNA seq) of mouse gonad”.

(8) Line 181 - Figure 4E and 4I should be 3E and 3I.

We have now updated the figure reference in the text to correct one.

(9) Line 183 - missing period.

Added.

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